

Global Machine Learning Model Testing

Notebooks 5 and 7 forecast the **total** monthly arrivals directly, with one model for one series (a *local* model). This notebook trains **one global model on a panel of related series** (access route × arrival region × origin subdivision) and asks:

Does one global model trained on the components, **summed back up** to the total, forecast the total better than the direct models of notebooks 5 and 7?

A global model learns the patterns shared by the series (seasonality, the pandemic collapse and rebound, the recent growth) from many more rows than a single series has, while static features tell it which series each row belongs to (Joseph & Tackes, ch. 10).

The protocol is the one of notebook 7, so the results are directly comparable:

- The same **validation origins** (two before the pandemic, the two most recent years after it), each forecasting the next 12 months. The latest origin forecasts `val.parquet`.
- The same **selection metric**: the mean seasonal MASE over the origins, computed on the **total**. Here the total is the sum of the component forecasts (**bottom-up**); the forecast of the total series by the same global model (**direct**) and the average MASE of the individual series are reported alongside.
- The same families (Ridge, Lasso, LightGBM, XGBoost), strategies (recursive, direct with a single model), COVID options (`keep` , `remove`), baselines and Optuna tuning.
- The **test set is never read here**.

Choice	Options
Model family	Ridge, Lasso, LightGBM, XGBoost
Target processing (per series)	<code>log</code> , <code>seasonal_log_diff</code> , <code>mean_scale</code>
Training window	all history, the last 10 years, the last 6 years
COVID rows	<code>keep</code> , or <code>remove</code> 2020–2022 from every series
Multi-step strategy	recursive, direct with a single model
Feature set	three combinations of lags, growth, rolling statistics, domain flags, calendar, always with the static series features

Only the configurations whose mean seasonal MASE is **below 1.5** are logged to MLflow.

Setup

```

In [1]: import itertools
import json
import logging
import math
import os
import sys
import time
import warnings
from concurrent.futures import ThreadPoolExecutor
from typing import NamedTuple

import matplotlib.pyplot as plt
import mlflow
import numpy as np
import optuna
import pandas as pd
import seaborn as sns
from category_encoders import OneHotEncoder
from dotenv import load_dotenv
from IPython.display import display
from joblib import Parallel, delayed, parallel_config
from lightgbm import LGBMRegressor, early_stopping
from sklearn.linear_model import LassoCV, RidgeCV
from sklearn.metrics import r2_score
from sklearn.model_selection import TimeSeriesSplit
from tqdm import tqdm
from xgboost import XGBRegressor

load_dotenv()

ROOT_PATH = os.path.dirname(os.getcwd())
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')

COVID_START = pd.Timestamp('2020-04-01') # first pandemic month: April 2020
COVID_END = pd.Timestamp('2021-12-31')
POST_COVID_START = pd.Timestamp('2022-01-01')
COVID_RECOVERY_END = pd.Timestamp('2022-06-30')

TARGET = 'arrival_count' # original scale (arrivals)
Y = 'y' # model scale (after the target processing)
TS_ID = 'series_id'
FREQ = 'ME'
SEASONAL_PERIOD = 12
RANDOM_STATE = 42

# physical cores: the fits are CPU-bound
N_JOBS = max(1, (os.cpu_count() or 2) // 2)

sns.set_palette('colorblind')
pd.set_option('display.max_columns', 40)
pd.set_option('display.width', 200)
logging.getLogger('mlflow').setLevel(logging.ERROR)
warnings.filterwarnings('ignore', category=FutureWarning)
mlflow.set_tracking_uri(os.getenv('MLFLOW_TRACKING_URI'))

```

```
EXPERIMENT_NAME = 'Brazil Tourism Forecasting ML Global'
experiment = mlflow.set_experiment(EXPERIMENT_NAME)
```

Book Helpers

The training and feature helpers come from the companion repository of:

Joseph, M. & Tackes, J. *Modern Time Series Forecasting with Python*, 2nd ed.
(Packt, 2024) — [PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E](#), MIT License

Helper	Module
FeatureConfig , MissingValueConfig , ModelConfig	src/forecasting/ml_forecasting.py
MLForecast	src/forecasting/ml_forecasting.py
calculate_metrics	src/forecasting/ml_forecasting.py
add_lags , add_rolling_features , add_seasonal_rolling_features	src/feature_engineering/autoregressive_features.py
add_temporal_features , add_fourier_features	src/feature_engineering/temporal_features.py

Everything noted in notebook 7 still applies (`MLForecastFixed` , sorted feature columns, the `evaluate_model` fix, `add_fourier_features(max_value=12)`). Two more points matter on a panel:

- **EWMA per series.** With `ts_id` , the book's `add_ewma` shifts each series but then runs `.ewm()` over the whole column, so the first months of a series continue the EWMA of

the previous one. `add_ewma_per_series` below computes the same columns (same names) inside each series.

- **Contiguous series.** `add_rolling_features` with `ts_id` also rolls over the whole column; it is only correct because the per-series shift leaves `NaN` at the start of every series, which blocks the window. That requires each series' rows to be contiguous, so the feature builder always sorts the panel by series and date first.
- `FeatureConfig(index_cols=['date', 'series_id'])`: the rows are indexed by month **and** series, so every prediction stays attached to its series.

```
In [2]: # Helpers credited to: Joseph, M. & Tackes, J. "Modern Time Series Forecasting with
# Python", 2nd ed. (Packt, 2024) – MIT License
# github.com/PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E
MTSF_REPO_PATH = os.getenv(
    'MTSF_REPO_PATH',
    os.path.join(
        os.path.dirname(ROOT_PATH), 'Modern-Time-Series-Forecasting-with-Python-2E'
    ),
)
if MTSF_REPO_PATH not in sys.path:
    sys.path.insert(0, MTSF_REPO_PATH)

from src.feature_engineering.autoregressive_features import ( # noqa: E402
    add_lags,
    add_rolling_features,
    add_seasonal_rolling_features,
)
from src.feature_engineering.temporal_features import ( # noqa: E402
    add_fourier_features,
    add_temporal_features,
)
from src.forecasting.ml_forecasting import ( # noqa: E402
    FeatureConfig,
    MissingValueConfig,
    MLForecast,
    ModelConfig,
    calculate_metrics,
)
```

1. The Panel of Series

The panel is built from the rows of `train.parquet` and `val.parquet` (the test set is never read). Every rule below is computed from the **training** rows, so the panel follows the data when it is updated:

Rule		Value	Why
Access routes	Aérea, Terrestre		The two routes carry almost all arrivals; river and sea are tiny and sparse

Rule	Value	Why
Origin subdivisions	the <code>N_SUBDIVISIONS</code> = 3 largest by training volume	Today LATAM, Europe and Anglo-America, about 90% of the arrivals
Arrival regions	the five real regions	<code>Outros</code> (unknown states) is tiny and irregular
Minimum volume	mean of the last 12 training months $\geq$ <code>MIN_MONTHLY_ARRIVALS</code> = 1,000	Tiny series add noise, not signal

Combinations that never occur (land arrivals into the Nordeste and Sudeste, which have no land border) are dropped by the volume rule. Two more series complete the panel:

- **remainder** = total – sum of the kept components (the other routes, subdivisions and regions, and the dropped small series). With it, the sum of the components **reconstructs the total exactly**, so a bottom-up forecast is a forecast of the same quantity as notebooks 5 and 7.
- **total**: the series of notebooks 5 and 7, so the same global model also forecasts it directly.

Every series gets four **static** features: `route`, `region`, `subdivision` and `level` (component, remainder or total). They are how a global model tells the series apart.

```
In [3]: ROUTES = {'Aérea': 'air', 'Terrestre': 'land'}
N_SUBDIVISIONS = 3
EXCLUDED_REGIONS = ['Outros']
MIN_MONTHLY_ARRIVALS = 1000
STATIC = ['route', 'region', 'subdivision', 'level']
SUBDIVISION_LABELS = {'Western / Eastern Europe': 'Europe'}

def load_rows(file_name):
    return pd.read_parquet(os.path.join(DATA_FOLDER_PATH, file_name))

train_rows = load_rows('train.parquet')
rows = pd.concat([train_rows, load_rows('val.parquet')], ignore_index=True)
TRAIN_END = train_rows['date'].max()
dates = pd.date_range(rows['date'].min(), rows['date'].max(), freq=FREQ)

top_subdivisions = (
    train_rows
    .groupby('country_cultural_subdivision')[TARGET]
    .sum()
    .nlargest(N_SUBDIVISIONS)
    .index.tolist()
)
candidates = rows[
    rows['access_route'].isin(ROUTES)
    & rows['country_cultural_subdivision'].isin(top_subdivisions)
    & ~rows['state_region'].isin(EXCLUDED_REGIONS)
```

```

]
components = (
    candidates
    .pivot_table(
        index=['access_route', 'state_region', 'country_cultural_subdivision'],
        columns='date',
        values=TARGET,
        aggfunc='sum',
    )
    .reindex(columns=dates, fill_value=0)
    .fillna(0)
)
recent_mean = components.loc[:, :TRAIN_END].iloc[:, -12:].mean(axis=1)
components = components[recent_mean >= MIN_MONTHLY_ARRIVALS]

total = rows.groupby('date')[TARGET].sum().reindex(dates)
remainder = total - components.sum()

def component_static(route, region, subdivision):
    return {
        'route': ROUTES[route],
        'region': region.replace('Região ', ''),
        'subdivision': SUBDIVISION_LABELS.get(subdivision, subdivision),
        'level': 'component',
    }

series_static = {}
series_values = {}
for key, values in components.iterrows():
    static = component_static(*key)
    series_id = f'{static["route"]}.{static["region"]}.{static["subdivision"]}'
    series_static[series_id] = static
    series_values[series_id] = values
for series_id, values in [('remainder', remainder), ('total', total)]:
    series_static[series_id] = {col: series_id for col in STATIC[:-1]} | {
        'level': series_id
    }
    series_values[series_id] = values

SERIES = list(series_values)
COMPONENTS = [s for s in SERIES if series_static[s]['level'] == 'component']
BOTTOM_UP = [*COMPONENTS, 'remainder'] # their sum is the total
STATIC_CATEGORIES = {
    col: sorted({static[col] for static in series_static.values()}) for col in STAT
}

# one row per (series, month), contiguous per series
panel = pd.concat(
    [
        pd.DataFrame({'date': dates, TS_ID: series_id, TARGET: values.to_numpy()})
        for series_id, values in series_values.items()
    ],
    ignore_index=True,
)

```

```

panel_wide = panel.pivot_table(
    index='date', columns=TS_ID, values=TARGET, aggfunc='sum'
)[SERIES]
assert np.allclose(panel_wide[BOTTOM_UP].sum(axis=1), panel_wide['total'])

train_share = panel_wide.loc[:TRAIN_END]
summary = pd.DataFrame({
    'level': [series_static[s]['level'] for s in SERIES],
    'share of total (%)': (train_share.sum() / train_share['total'].sum() * 100).round(1),
    'mean last 12 train months': train_share.iloc[-12:].mean().round(0),
    'zero months': (panel_wide == 0).sum(),
    'zero months since 2022': (panel_wide.loc[POST_COVID_START:] == 0).sum(),
})
print(
    f'Subdivisions: {top_subdivisions} | {len(COMPONENTS)} components + remainder + '
    f'total = {len(SERIES)} series x {len(dates)} months = {len(panel):,} rows'
)
print(
    f'Components cover {summary.loc[COMPONENTS, "share of total (%)"].sum():.1f}% of '
    'the training arrivals'
)
summary

```

Subdivisions: ['LATAM', 'Western / Eastern Europe', 'Anglo-America'] | 19 components + remainder + total = 21 series x 440 months = 9,240 rows
 Components cover 86.2% of the training arrivals

Out[3]:

	level	share of total (%)	mean last 12 train months	zero months	zero months since 2022
series_id					
air-Centro-Oeste-Anglo-America	component	0.2	1686.0	101	0
air-Centro-Oeste-LATAM	component	0.2	1945.0	79	0
air-Centro-Oeste-Europe	component	0.2	1468.0	115	0
air-Nordeste-Anglo-America	component	0.3	1896.0	5	0
air-Nordeste-LATAM	component	1.0	7668.0	6	0
air-Nordeste-Europe	component	3.6	13143.0	3	0
air-Norte-Anglo-America	component	0.3	1510.0	5	0
air-Norte-LATAM	component	0.3	1236.0	3	0
air-Norte-Europe	component	0.2	1375.0	6	0
air-Sudeste-Anglo-America	component	13.3	68046.0	0	0
air-Sudeste-LATAM	component	19.4	131873.0	0	0
air-Sudeste-Europe	component	16.2	65653.0	0	0
air-Sul-LATAM	component	2.8	23602.0	4	0
land-Centro-Oeste-LATAM	component	1.2	6222.0	0	0
land-Norte-LATAM	component	0.4	2734.0	17	0
land-Norte-Europe	component	0.1	2152.0	29	0
land-Sul-Anglo-America	component	0.7	5139.0	3	0
land-Sul-LATAM	component	24.2	127278.0	0	0
land-Sul-Europe	component	1.6	7762.0	0	0
remainder	remainder	13.9	55958.0	0	0
total	total	100.0	528344.0	0	0

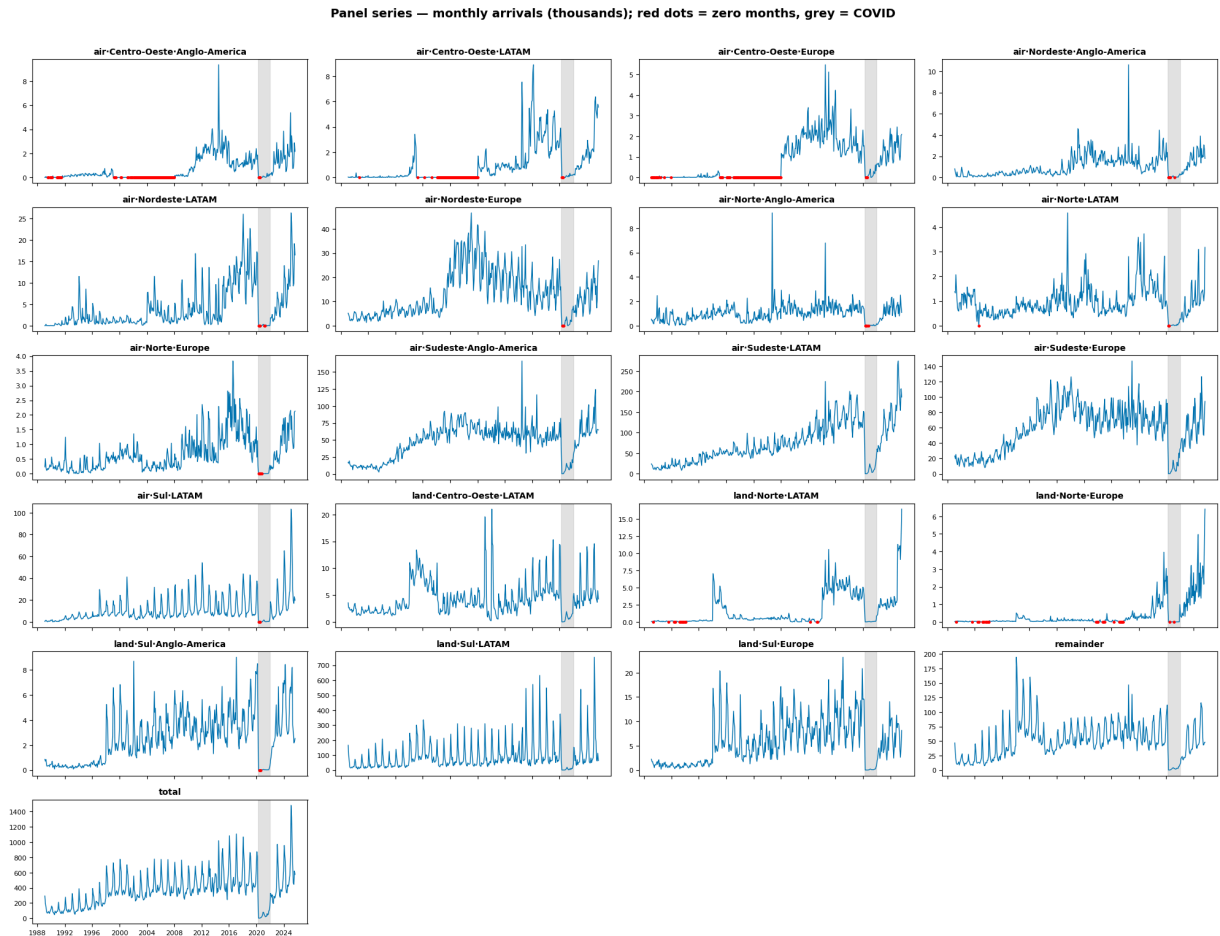
```
In [4]: N_COLS = 4
n_rows = math.ceil(len(SERIES) / N_COLS)
fig, axes = plt.subplots(n_rows, N_COLS, figsize=(20, 2.6 * n_rows), sharex=True)
for ax, series_id in zip(axes.flat, SERIES):
    values = panel_wide[series_id]
    ax.plot(values.index, values / 1e3, linewidth=1)
    zeros = values[values == 0]
    ax.scatter(zeros.index, zeros, color='red', s=6, zorder=3)
```



```

ax.axvspan(COVID_START, COVID_END, color='grey', alpha=0.2)
ax.set_title(series_id, fontsize=10, fontweight='bold')
ax.tick_params(labelsize=8)
for ax in axes.flat[len(SERIES) :]:
    ax.set_visible(False)
fig.suptitle(
    'Panel series — monthly arrivals (thousands); red dots = zero months, grey = CO
    fontsize=14,
    fontweight='bold',
)
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()

```



2. Validation Origins, Windows, COVID Rows and Target Processing

Origins. The same rule as notebooks 5 and 7: the end of training (its window is `val.parquet`) and one year earlier, plus the two most recent yearly origins whose 12-month validation window ends before the pandemic. They are counted back from the end of training, so they roll forward with the data. Each origin only sees the history up to that origin.

Windows (counted back from the origin): `all`, `last_10y`, `last_6y`. `post_covid` is left out: it is short for a panel with long histories and only exists at two origins.

COVID rows (as in notebook 7): `keep`, or `remove` the years 2020–2022 from **every** series before the target processing. At the pre-COVID origins `remove` is the same model as `keep`.

Target processing, fitted **per series** on its own history at each origin. The series differ by orders of magnitude (the total is about 30 times the smallest component), so a global model needs every series on a comparable scale; `none` would let the largest series dominate the loss.

Option	Transform (per series)	Inverse
<code>log</code>	$\log(1 + y)$	$\exp(\hat{y}) - 1$
<code>seasonal_log_diff</code>	$\log(1 + y_t) - \log(1 + y_{\{t-12\}})$, the yearly growth (notebook 7)	$(1 + y_{\{t-12\}}) \cdot \exp(\hat{z}) - 1$
<code>mean_scale</code>	y / m , m = mean of the series' last 12 months	$\hat{y} \cdot m$

`log(1 + y)` instead of `log(y)` because some components have zero months in the early years. `auto_stationary` is left out: it would be one statistical pipeline per series, and its tests are fragile on the small series.

```
In [5]: HORIZON = 12
N_PRE_COVID_ORIGINS = 2
N_POST_COVID_ORIGINS = 2

def validation_window(origin):
    """First and last month forecast from `origin` (the last month the model sees).
    return origin + pd.offsets.MonthEnd(1), origin + pd.offsets.MonthEnd(HORIZON)

n_train_months = len(panel_wide.loc[:TRAIN_END])
candidates = [TRAIN_END] + [
    TRAIN_END - pd.offsets.MonthEnd(12 * k) for k in range(1, n_train_months // 12)
]
post_covid_origins = [o for o in candidates if validation_window(o)[0] > COVID_END]
                    :N_POST_COVID_ORIGINS
]
pre_covid_origins = [o for o in candidates if validation_window(o)[1] < COVID_START]
                    :N_PRE_COVID_ORIGINS
]
ORIGINS = sorted(pre_covid_origins + post_covid_origins)
LATEST_ORIGIN = TRAIN_END # its validation window is val.parquet
assert LATEST_ORIGIN in ORIGINS
assert validation_window(LATEST_ORIGIN)[1] == dates[-1]

def origin_label(origin):
    return f'{origin:%Y-%m}'
```

```

def val_dates_at(origin):
    start, end = validation_window(origin)
    return panel_wide.loc[start:end].index

WINDOWS = {'all': None, 'last_10y': 10, 'last_6y': 6}
REMOVED_START = pd.Timestamp('2020-01-31')
REMOVED_END = pd.Timestamp('2022-12-31')
COVID_ROWS = ['keep', 'remove']

def window_start(window, origin):
    years = WINDOWS[window]
    return dates[0] if years is None else origin - pd.DateOffset(years=years)

def effective_covid_rows(covid_rows, origin):
    """Before the pandemic there is nothing to remove: `remove` equals `keep`."""
    return 'keep' if origin < REMOVED_START else covid_rows

def training_panel(covid_rows, origin):
    """The panel up to the origin, without the removed years for `remove`."""
    part = panel[panel['date'] <= origin]
    if covid_rows == 'remove':
        part = part[~part['date'].between(REMOVED_START, REMOVED_END)]
    return part.reset_index(drop=True)

pd.DataFrame([
    {
        'origin': origin_label(origin),
        'phase': 'pre-COVID' if origin < COVID_START else 'post-COVID',
        'validation window': f'{start:%b %Y} → {end:%b %Y}',
        'history months': len(panel_wide.loc[:origin]),
    }
    for origin in ORIGINS
    for start, end in [validation_window(origin)]
])

```

Out[5]:

	origin	phase	validation window	history months
0	2017-08	pre-COVID	Sep 2017 → Aug 2018	344
1	2018-08	pre-COVID	Sep 2018 → Aug 2019	356
2	2023-08	post-COVID	Sep 2023 → Aug 2024	416
3	2024-08	post-COVID	Sep 2024 → Aug 2025	428

In [6]: TARGET_TRANSFORMS = ['log', 'seasonal_log_diff', 'mean_scale']

```

class Log1pTarget:
    """log(1 + y); inverse exp(z) - 1."""

```

```

@staticmethod
def transform(y):
    return np.log1p(y.astype(float))

@staticmethod
def inverse_transform(z):
    return np.expm1(np.asarray(z, dtype=float))

class SeasonalLog1pDiffTarget:
    """z_t = log(1 + y_t) - log(1 + y_{t-12}), positional (notebook 7, with log1p).

    The inverse takes y_{t-12} from the last `period` months of the fitted history,
    it is exact for forecasts up to `period` months after the end of that history.
    """

    def __init__(self, history, period=SEASONAL_PERIOD):
        self.period = period
        self.log_history = np.log1p(history.astype(float)).to_numpy()
        self.last_date = history.index[-1]

    def transform(self, y):
        log_y = np.log1p(y.astype(float))
        return log_y - log_y.shift(self.period)

    def inverse_transform(self, z, dates):
        dates = pd.DatetimeIndex(dates)
        steps = (dates.year - self.last_date.year) * 12 + (
            dates.month - self.last_date.month
        )
        assert ((steps >= 1) & (steps <= self.period)).all(), 'up to one season ahead'
        base = self.log_history[len(self.log_history) - self.period + steps - 1]
        return np.expm1(np.asarray(z, dtype=float) + base)

class MeanScaleTarget:
    """y / m, with m the mean of the last 12 months of the fitted history."""

    def __init__(self, history):
        self.scale = max(float(history.iloc[-SEASONAL_PERIOD:].mean()), 1.0)

    def transform(self, y):
        return y.astype(float) / self.scale

    def inverse_transform(self, z):
        return np.asarray(z, dtype=float) * self.scale

def fit_target_transform(name, history):
    if name == 'log':
        return Log1pTarget()
    if name == 'seasonal_log_diff':
        return SeasonalLog1pDiffTarget(history)
    if name == 'mean_scale':
        return MeanScaleTarget(history)
    raise ValueError(f'unknown target transform: {name}')

```

```

def inverse(transformer, z, dates):
    if isinstance(transformer, SeasonalLog1pDiffTarget):
        return transformer.inverse_transform(z, dates)
    return transformer.inverse_transform(z)

def make_setup(transform_name, covid_rows, origin):
    """Per-series transformers and the model-scale panel history of one origin."""
    part = training_panel(covid_rows, origin)
    transformers, history = {}, []
    for series_id, frame in part.groupby(TS_ID, sort=False):
        values = frame.set_index('date')[TARGET]
        transformers[series_id] = fit_target_transform(transform_name, values)
        history.append(
            frame.assign(**{Y: transformers[series_id].transform(values).to_numpy()
            })
    return {
        'origin': origin,
        'transform': transform_name,
        'covid_rows': covid_rows,
        'transformers': transformers,
        'history': pd.concat(history, ignore_index=True)[['date', TS_ID, Y, TARGET]]
    }

setups = {
    (origin, name, covid_rows): make_setup(name, covid_rows, origin)
    for origin in ORIGINS
    for name in TARGET_TRANSFORMS
    for covid_rows in COVID_ROWS
    if effective_covid_rows(covid_rows, origin) == covid_rows
}
print(f'{len(setups)} setups (origin x target processing x COVID rows)')

```

18 setups (origin x target processing x COVID rows)

3. Feature Builder for the Panel

`build_features(df, shift)` is notebook 7's builder applied **per series**: the same groups, the same parameters (`feature_groups.json` , notebook 6) and the same `shift` logic (`shift = 1` for the recursive model, `shift = 12` for the direct single model). What changes on a panel:

Change	Why
The panel is sorted by series and date first	The book's rolling helper needs contiguous series (see <i>Book Helpers</i>)
<code>growth</code> is computed within each series	A plain <code>shift(1)</code> would divide the first month of a series by the last month of the previous one

Change	Why
<code>add_ewma_per_series</code> replaces <code>add_ewma</code>	Same reason, for the EWMA
New group <code>static : route , region , subdivision , level</code> (categorical)	Tells the global model which series a row belongs to
Zero months give an undefined growth (division by zero), set to <code>NaN</code>	Only in the early years of a few components; those rows are dropped from training like the warm-up rows

```
In [7]: with open(os.path.join(DATA_FOLDER_PATH, 'feature_groups.json'), encoding='utf-8')
        FEATURE_SPEC = json.load(f)
        PARAMS = FEATURE_SPEC['parameters']
        SEASONAL_LAGS = [
            lag
            for lag in FEATURE_SPEC['strategies']['recursive']['lags']
            if lag >= SEASONAL_PERIOD
        ]

        GROUP_ORDER = [
            'lags',
            'growth',
            'rolling',
            'domain',
            'static',
            'cal_cat',
            'cal_fourier',
        ]
        TARGET_DERIVED = ['lags', 'growth', 'rolling']
        CATEGORICAL_GROUPS = ['static', 'cal_cat']

    def lags_for_shift(shift):
        """The two most recent legal lags plus the seasonal lags that are still legal."""
        return sorted({shift, shift + 1} | {lag for lag in SEASONAL_LAGS if lag >= shif

    def growth_lags_for_shift(shift):
        return sorted({shift} | ({SEASONAL_PERIOD} if SEASONAL_PERIOD >= shift else set

    def add_ewma_per_series(df, column, spans, n_shift):
        """`add_ewma` (book) computed inside each series, with the book's column names.
        by_series = df.groupby(TS_ID)[column]
        added = {
            f'{column}_ewma_span_{span}': by_series.transform(
                lambda x, span=span: x.shift(n_shift).ewm(span=span, adjust=False).mean
            )
            for span in spans
        }
        return df.assign(**added), list(added)
```

```

def is_spliced(df):
    """True when the removed COVID years are missing from the history."""
    return (
        not df['date'].between(REMOVED_START, REMOVED_END).any()
        and (df['date'] > REMOVED_END).any()
    )

def add_domain_features(df, lags):
    df['days_in_month'] = df['date'].dt.days_in_month
    if is_spliced(df):
        # a lag crosses the gap when the row is after it and the lagged row before
        df['is_pre_gap'] = (df['date'] < REMOVED_START).astype(int)
        df, pre_gap_lags = add_lags(df, lags=lags, column='is_pre_gap', ts_id=TS_ID)
        after_gap = (df['date'] > REMOVED_END).astype(int)
        flags = []
        for lag, column in zip(lags, pre_gap_lags):
            flag = f'crosses_gap_lag_{lag}'
            df[flag] = (df[column].fillna(1) * after_gap).astype(int)
            flags.append(flag)
        return df, ['days_in_month', *flags]
    df['is_covid'] = df['date'].between(COVID_START, COVID_END).astype(int)
    df['is_covid_recovery'] = (
        df['date'].between(POST_COVID_START, COVID_RECOVERY_END).astype(int)
    )
    df, covid_lags = add_lags(df, lags=lags, column='is_covid', ts_id=TS_ID)
    df[covid_lags] = df[covid_lags].fillna(0).astype(int)
    return df, ['is_covid', 'is_covid_recovery', 'days_in_month', *covid_lags]

def add_static_features(df):
    for col in STATIC:
        values = df[TS_ID].map(lambda s, col=col: series_static[s][col])
        df[col] = pd.Categorical(values, categories=STATIC_CATEGORIES[col])
    return df, list(STATIC)

def build_features(df, shift):
    """Every feature group, per series, for a model that knows the target `shift` b

    `df` has `date`, `series_id`, `y` (model scale) and `arrival_count` (arrivals).
    Future rows have NaN targets. Returns the feature table (sorted by series and d
    and {group: [columns]}).
    """
    df = df.sort_values([TS_ID, 'date']).reset_index(drop=True)
    groups = {}
    lags = lags_for_shift(shift)

    # target-derived groups (book helpers, per series)
    df, groups['lags'] = add_lags(df, lags=lags, column=Y, ts_id=TS_ID)
    previous = df.groupby(TS_ID)[TARGET].shift(1)
    df['arrivals_mom_growth'] = (df[TARGET] / previous - 1).replace(
        [np.inf, -np.inf], np.nan
    )
    df, groups['growth'] = add_lags(
        df, lags=growth_lags_for_shift(shift), column='arrivals_mom_growth', ts_id=

```

```

)
df, rolling = add_rolling_features(
    df,
    rolls=PARAMS['rolling_windows'],
    column=Y,
    agg_funcs=PARAMS['rolling_aggs'],
    ts_id=TS_ID,
    n_shift=shift,
)
df, seasonal = add_seasonal_rolling_features(
    df,
    seasonal_periods=[PARAMS['seasonal_period']],
    rolls=PARAMS['seasonal_windows'],
    column=Y,
    agg_funcs=PARAMS['seasonal_aggs'],
    ts_id=TS_ID,
    n_shift=math.ceil(shift / PARAMS['seasonal_period']), # counted in seasons
)
df, ewma = add_ewma_per_series(df, Y, PARAMS['ewma_spans'], shift)
groups['rolling'] = rolling + seasonal + ewma

# calendar groups (book helpers)
df, _ = add_temporal_features(
    df,
    field_name='date',
    frequency=FREQ,
    add_elapsed=True,
    prefix='cal',
    drop=False,
)
df, groups['cal_fourier'] = add_fourier_features(
    df,
    column_to_encode='cal_Month',
    max_value=12,
    n_fourier_terms=PARAMS['fourier_terms'],
)
df['cal_Month'] = pd.Categorical(df['cal_Month'], categories=range(1, 13))
groups['cal_cat'] = ['cal_Month']

df, groups['domain'] = add_domain_features(df, lags)
df, groups['static'] = add_static_features(df)
return df, groups

def training_rows(features, groups, start, end):
    """Rows of the training window with a known target and complete target features
    target_derived = [c for g in TARGET_DERIVED for c in groups[g]]
    rows = features[features['date'].between(start, end)]
    return rows.dropna(subset=[Y, *target_derived])

example, example_groups = build_features(
    setups[LATEST_ORIGIN, 'log', 'keep']['history'], shift=1
)
display(
    pd.DataFrame({

```



```

        'n_columns': {g: len(example_groups[g]) for g in GROUP_ORDER},
        'columns': {g: ', '.join(example_groups[g][:6]) for g in GROUP_ORDER},
    })
)
example[example['date'] == LATEST_ORIGIN][
    ['date', TS_ID, TARGET, Y, f'{Y}_lag_12', f'{Y}_ewma_span_3', *STATIC]
].head(6)

```

	n_columns	columns
lags	6	y_lag_1, y_lag_2, y_lag_12, y_lag_13, y_lag_24...
growth	2	arrivals_mom_growth_lag_1, arrivals_mom_growth...
rolling	19	y_rolling_3_mean, y_rolling_3_std, y_rolling_3...
domain	9	is_covid, is_covid_recovery, days_in_month, is...
static	4	route, region, subdivision, level
cal_cat	1	cal_Month
cal_fourier	6	cal_Month_sin_1, cal_Month_sin_2, cal_Month_si...

Out[7]:

	date	series_id	arrival_count	y	y_lag_12	y_ewma_span_3	route
427	2024-08-31	air-Centro-Oeste-Anglo-America	2128.0	7.663408	7.611348	7.325877	air
855	2024-08-31	air-Centro-Oeste-Europe	1747.0	7.466228	7.066467	7.094649	air
1283	2024-08-31	air-Centro-Oeste-LATAM	2176.0	7.685703	7.337588	7.399641	air
1711	2024-08-31	air-Nordeste-Anglo-America	2678.0	7.893199	7.727094	7.760078	air
2139	2024-08-31	air-Nordeste-Europe	19346.0	9.870293	9.604610	9.203143	air
2567	2024-08-31	air-Nordeste-LATAM	8495.0	9.047351	8.453188	8.909425	air

Feature Sets, Feature Config and Missing Values

Three feature sets, all with the `static` group (without it the global model cannot tell the series apart). They vary the calendar encoding and the month-over-month growth, the choices that notebook 7 left open. The elapsed-time feature is left out: it describes the calendar, not the series, and trees cannot extrapolate it.

Each feature set becomes a `FeatureConfig` (book) with `index_cols=['date', 'series_id']`. The static features and `cal_Month` are categorical; everything else is

continuous. The `MissingValueConfig` back-fills the target-derived columns, as in notebook 7.

```
In [8]: FEATURE_SETS = [
    ('lags', 'rolling', 'domain', 'static', 'cal_fourier'),
    ('lags', 'rolling', 'domain', 'static', 'cal_cat'),
    ('lags', 'growth', 'rolling', 'domain', 'static', 'cal_fourier'),
]

def canonical(feature_set):
    """Order the groups of a feature set consistently."""
    return tuple(g for g in GROUP_ORDER if g in feature_set)

def set_label(feature_set):
    return '+'.join(canonical(feature_set))

def categorical_columns(groups, feature_set):
    return [col for g in CATEGORICAL_GROUPS if g in feature_set for col in groups[g]]

def make_feature_config(groups, feature_set):
    continuous = [
        col
        for g in canonical(feature_set)
        if g not in CATEGORICAL_GROUPS
        for col in groups[g]
    ]
    return FeatureConfig(
        date='date',
        target=Y,
        original_target=TARGET,
        continuous_features=continuous,
        categorical_features=categorical_columns(groups, feature_set),
        index_cols=['date', TS_ID],
    )

def make_missing_config(groups):
    return MissingValueConfig(
        bfill_columns=groups['lags'] + groups['growth'] + groups['rolling']
    )
```

4. Model Families

The four families of notebook 7, created fresh for every fit. On the panel the training rows are sorted **by month** (all series of a month together) before fitting, so that:

- the **temporal cross-validation** of Ridge and Lasso (`TimeSeriesSplit`) splits by time, with validation folds of about 12 months of **every** series;

- the **early stopping** of the trees holds out the last 12 months of every series.

Family	Estimator	Static and month features
ridge	RidgeCV(alphas=10 ⁻³ ... 10 ³ , cv=TimeSeriesSplit)	one-hot
lasso	LassoCV(30 alphas, cv=TimeSeriesSplit)	one-hot
lightgbm	LGBMRegressor (num_leaves=8 , early stopping)	native category
xgboost	XGBRegressor (max_depth=3 , early stopping)	native category

The panel has about 20 times more rows than the total series, which is the point of a global model; the tree parameters are those of notebook 7 and are tuned with Optuna in section 8.

```
In [9]: FAMILIES = ['ridge', 'lasso', 'lightgbm', 'xgboost']
FAMILY_NAMES = {
    'ridge': 'Ridge',
    'lasso': 'Lasso',
    'lightgbm': 'LightGBM',
    'xgboost': 'XGBoost',
}
TREE_FAMILIES = {'lightgbm', 'xgboost'}

CV_SPLITS = 5
CV_TEST_MONTHS = 12
RIDGE_ALPHAS = np.logspace(-3, 3, 13)

N_TREES = 2000
LGBM_PARAMS = {
    'n_estimators': N_TREES,
    'learning_rate': 0.02,
    'num_leaves': 8,
    'min_child_samples': 10,
    'subsample': 0.8,
    'subsample_freq': 1,
    'colsample_bytree': 0.8,
    'random_state': RANDOM_STATE,
    'n_jobs': 1,
    'verbose': -1,
}
XGB_PARAMS = {
    'n_estimators': N_TREES,
    'learning_rate': 0.02,
    'max_depth': 3,
    'min_child_weight': 3,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'tree_method': 'hist',
    'enable_categorical': True,
    'random_state': RANDOM_STATE,
    'n_jobs': 1,
}
```

```

EARLY_STOPPING_ROUNDS = 100
EARLY_STOPPING_MONTHS = 12 # the last 12 months of every series in the window
MIN_MONTHS_FOR_EARLY_STOPPING = 3 * EARLY_STOPPING_MONTHS
SHORT_WINDOW_TREES = 300

class MLForecastFixed(MLForecast):
    """Book MLForecast with the predict-time column check on the *input* columns."""

    def fit(self, X, y, is_transformed=False, fit_kwargs=None):
        self._input_features = X.columns.tolist()
        return super().fit(
            X, y, is_transformed=is_transformed, fit_kwargs=fit_kwargs or {}
        )

    def predict(self, X):
        missing = set(self._input_features) - set(X.columns)
        assert not missing, f'features missing at prediction time: {missing}'
        X = X[self._input_features].copy()
        if self.model_config.fill_missing:
            X = self.missing_config.impute_missing_values(X)
        if self.model_config.encode_categorical:
            X = self._cat_encoder.transform(X)
        if self.model_config.normalize:
            scaled = self._continuous_feats + self._encoded_categorical_features
            X[scaled] = self._scaler.transform(X[scaled])
        return pd.Series(
            self._model.predict(X[self._train_features]).ravel(),
            index=X.index,
            name=f'{self.model_config.name}',
        )

    def temporal_cv(n_months, n_series):
        """TimeSeriesSplit on month-sorted rows: folds of ~12 months of every series."""
        n_splits = max(2, min(CV_SPLITS, n_months // CV_TEST_MONTHS - 1))
        test_months = min(CV_TEST_MONTHS, n_months // (n_splits + 1))
        return TimeSeriesSplit(n_splits=n_splits, test_size=test_months * n_series)

    def make_estimator(family, n_months, n_series, tree_params=None):
        if family == 'ridge':
            return RidgeCV(alphas=RIDGE_ALPHAS, cv=temporal_cv(n_months, n_series))
        if family == 'lasso':
            return LassoCV(alphas=30, cv=temporal_cv(n_months, n_series), max_iter=2000)
        if family == 'lightgbm':
            return LGBMRegressor(**{**LGBM_PARAMS, **(tree_params or {})})
        return XGBRegressor(**{**XGB_PARAMS, **(tree_params or {})})

    def make_model_config(family, categorical, shape, tree_params=None): # noqa: PLR09
        """A fresh ModelConfig (book) for one fit; `shape` = (months, series)."""
        is_tree = family in TREE_FAMILIES
        encode = bool(categorical) and not is_tree
        return ModelConfig(
            model=make_estimator(family, *shape, tree_params),

```

```

        name=FAMILY_NAMES[family],
        normalize=not is_tree,
        fill_missing=not is_tree,
        encode_categorical=encode,
        categorical_encoder=OneHotEncoder(cols=categorical) if encode else None,
    )

def feature_matrix(feature_config, rows):
    """`get_X_y` (book) with the columns in a fixed, sorted order."""
    X, y, _ = feature_config.get_X_y(rows, categorical=True)
    return X[sorted(X.columns)], y

def best_n_estimators( # noqa: PLR0913, PLR0917
    family, feature_config, missing_config, X, y, tree_params
):
    """Early stopping on the last 12 months of every series (never on validation)."""
    months = X.index.get_level_values('date')
    holdout = months > np.sort(months.unique())[-EARLY_STOPPING_MONTHS - 1]
    shape = (months[~holdout].nunique(), X.index.get_level_values(TS_ID).nunique())
    probe_config = make_model_config(
        family, feature_config.categorical_features, shape, tree_params
    )
    eval_set = [(X[holdout], y[holdout])]
    if family == 'lightgbm':
        fit_kwargs = {
            'eval_set': eval_set,
            'callbacks': [early_stopping(EARLY_STOPPING_ROUNDS, verbose=False)],
        }
    else:
        probe_config.model.set_params(early_stopping_rounds=EARLY_STOPPING_ROUNDS)
        fit_kwargs = {'eval_set': eval_set, 'verbose': False}
    probe = MLForecastFixed(
        model_config=probe_config,
        feature_config=feature_config,
        missing_config=missing_config,
    )
    probe.fit(X[~holdout], y[~holdout], fit_kwargs=fit_kwargs)
    if family == 'lightgbm':
        return probe._model.best_iteration_ or N_TREES
    return probe._model.best_iteration + 1

def fit_model(family, feature_config, missing_config, train, tree_params=None):
    """Chapter 8's `evaluate_model` (fit part) on month-sorted panel rows."""
    train = train.sort_values(['date', TS_ID])
    X, y = feature_matrix(feature_config, train)
    y = y[Y]
    shape = (train['date'].nunique(), train[TS_ID].nunique())
    model_config = make_model_config(
        family, feature_config.categorical_features, shape, tree_params
    )
    if family in TREE_FAMILIES:
        n_trees = (
            best_n_estimators(family, feature_config, missing_config, X, y, tree_pa

```

```

        if shape[0] >= MIN_MONTHS_FOR_EARLY_STOPPING
        else SHORT_WINDOW_TREES
    )
    model_config.model.set_params(n_estimators=n_trees)
model = MLForecastFixed(
    model_config=model_config,
    feature_config=feature_config,
    missing_config=missing_config,
)
model.fit(X, y)
return model

def model_info(family, model):
    """Tree count, or chosen penalty and whether it sits on the edge of its grid."""
    estimator = model._model
    if family in TREE_FAMILIES:
        return {'n_trees': int(estimator.get_params()['n_estimators'])}
    grid = np.asarray(estimator.alphas if family == 'ridge' else estimator.alphas_)
    alpha = float(estimator.alpha_)
    return {
        'alpha': alpha,
        'alpha_at_edge': bool(
            np.isclose(alpha, grid.min()) or np.isclose(alpha, grid.max())
        ),
    }

def predict_rows(model, feature_config, rows):
    X, _ = feature_matrix(feature_config, rows)
    return model.predict(X)

```

5. Multi-step Strategies on the Panel

The two strategies of notebook 7, now over all series at once:

Strategy	How the 12 months are produced
recursive (<code>shift = 1</code>)	Predict <code>T+1</code> for every series, convert each prediction back to arrivals with its own series' inverse, append the month to the panel, rebuild the features, predict <code>T+2</code> , ... 12 times
direct_single (<code>shift = 12</code>)	One pass: every feature of the 12 future months is already known

Every forecast is returned as a table with one column per series (arrivals), so the bottom-up total is the sum of the component and remainder columns.

```
In [10]: STRATEGIES = ['recursive', 'direct_single']
```

```

class Config(NamedTuple):
    family: str

```

```

transform: str
window: str
covid_rows: str
strategy: str
feature_set: tuple

@property
def label(self):
    return (
        f'{FAMILY_NAMES[self.family]} | {self.transform} | {self.window} | '
        f'{self.covid_rows} | {self.strategy} | {set_label(self.feature_set)}'
    )

def to_arrivals(setup, predicted_z):
    """Inverse-transform a long frame (date, series_id, z) to arrivals, per series.
    out = predicted_z.copy()
    for series_id, frame in predicted_z.groupby(TS_ID):
        out.loc[frame.index, TARGET] = inverse(
            setup['transformers'][series_id], frame['z'], frame['date']
        )
    return out

def with_future_rows(history, future_dates):
    future = pd.DataFrame(
        [(d, s) for s in SERIES for d in future_dates], columns=['date', TS_ID]
    ).assign(**{Y: np.nan, TARGET: np.nan})
    return pd.concat([history, future], ignore_index=True)

def fit_for_shift(config, setup, frame, shift, tree_params=None): # noqa: PLR0917
    features, groups = build_features(frame, shift)
    feature_config = make_feature_config(groups, config.feature_set)
    origin = setup['origin']
    model = fit_model(
        config.family,
        feature_config,
        make_missing_config(groups),
        training_rows(features, groups, window_start(config.window, origin), origin,
        tree_params,
    )
    return model, feature_config, features

def predicted_frame(model, feature_config, rows):
    z = predict_rows(model, feature_config, rows)
    return pd.DataFrame({
        'date': z.index.get_level_values('date'),
        TS_ID: z.index.get_level_values(TS_ID),
        'z': z.to_numpy(),
    })

def forecast_recursive(config, setup, future_dates, tree_params=None):
    model, feature_config, _ = fit_for_shift(

```

```

        config, setup, setup['history'], 1, tree_params
    )
    extended = setup['history']
    steps = []
    for date in future_dates:
        extended = with_future_rows(extended, [date])
        features, _ = build_features(extended, 1)
        step = to_arrivals(
            setup,
            predicted_frame(model, feature_config, features[features['date'] == date])
        )
        # feed the predictions back: model scale and arrivals, per series
        fed = step.set_index(TS_ID)
        is_new = extended['date'] == date
        extended.loc[is_new, Y] = extended.loc[is_new, TS_ID].map(fed['z']).to_numpy()
        extended.loc[is_new, TARGET] = (
            extended.loc[is_new, TS_ID].map(fed[TARGET]).to_numpy()
        )
        steps.append(step)
    return pd.concat(steps, ignore_index=True), model

def forecast_direct_single(config, setup, future_dates, tree_params=None):
    extended = with_future_rows(setup['history'], future_dates)
    model, feature_config, features = fit_for_shift(
        config, setup, extended, len(future_dates), tree_params
    )
    rows = features[features['date'].isin(future_dates)]
    return to_arrivals(setup, predicted_frame(model, feature_config, rows)), model

FORECASTERS = {
    'recursive': forecast_recursive,
    'direct_single': forecast_direct_single,
}

```

Scoring One Configuration

At each origin the forecast table (one column per series) is scored in three views, all with the **seasonal MASE** of notebooks 5 and 7 (MAE divided by the in-sample MAE of the seasonal naive `y_{t-12}` on the series' history, pandemic pairs from April 2020 to December 2022 left out):

View	Forecast of the total	Role
bottom-up	sum of the component and remainder forecasts	Selection metric (<code>MASE_mean</code>) and all the total metrics (RMSE, MAE, MAPE, R^2 , Forecast Bias)
direct	the global model's forecast of the <code>total</code> series	<code>MASE_direct_mean</code>
series	—	<code>MASE_series_mean</code> : the average seasonal MASE of the components and the remainder

The bottom-up total and the total of notebooks 5 and 7 are the same series with the same MASE scale, so `MASE_mean` is directly comparable across the three notebooks. As in notebook 7, the origins of a configuration are combined into `MASE_mean` (ranking), `MASE_worst` and `MASE_post`, and a configuration that fails at any origin is left out of the ranking.

```
In [11]: METRICS = ['RMSE', 'MAE', 'MAPE', 'R2', 'MASE', 'Forecast Bias']
AGG_METRICS = [
    'MASE_mean',
    'MASE_worst',
    'MASE_post',
    'MASE_direct_mean',
    'MASE_series_mean',
    'n_origins',
]
# seasonal differences distorted by the pandemic collapse and rebound
MASE_EXCLUDED = (COVID_START, pd.Timestamp('2022-12-31'))

def seasonal_mase_scale(history):
    """In-sample MAE of the seasonal naive (y_t - y_{t-12}), pandemic pairs exclude
    diffs = (history - history.shift(SEASONAL_PERIOD)).abs()
    in_pandemic = history.index.to_series().between(*MASE_EXCLUDED)
    touches = in_pandemic | in_pandemic.shift(SEASONAL_PERIOD, fill_value=False)
    return float(diffs[~touches.to_numpy()].mean())

MASE_SCALES = {
    origin: {s: seasonal_mase_scale(panel_wide.loc[:origin, s]) for s in SERIES}
    for origin in ORIGINS
}

def score_total(actual, predicted, name, origin):
    metrics = calculate_metrics(
        actual, predicted, name
    ) # book: MAE, MSE, Forecast Bias
    errors = (actual - predicted).to_numpy()
    return {
        'RMSE': float(np.sqrt(metrics['MSE'])),
        'MAE': metrics['MAE'],
        'MAPE': float(np.mean(np.abs(errors / actual.to_numpy())) * 100),
        'R2': float(r2_score(actual.to_numpy(), predicted.to_numpy())),
        'MASE': float(metrics['MAE'] / MASE_SCALES[origin]['total']),
        'Forecast Bias': metrics['Forecast Bias'],
        'abs_errors': np.abs(errors),
    }

def score_forecast(forecast, name, origin):
    """Scores of one origin's forecast table (index = month, one column per series)
    actual = panel_wide.loc[forecast.index]
    series_mase = {
        s: float(np.mean(np.abs(actual[s] - forecast[s]))) / MASE_SCALES[origin][s])
```

```

        for s in SERIES
    }
    bottom_up = forecast[BOTTOM_UP].sum(axis=1)
    direct_mae = float(np.mean(np.abs(actual['total'] - forecast['total'])))
    return {
        **score_total(actual['total'], bottom_up, name, origin),
        'MASE_direct': direct_mae / MASE_SCALES[origin]['total'],
        'MASE_series': float(np.mean([series_mase[s] for s in BOTTOM_UP])),
        'series_mase': series_mase,
        'prediction': bottom_up.to_numpy(),
        'forecast': {s: forecast[s].to_numpy() for s in SERIES},
    }

def aggregate(per_origin):
    """Combine the per-origin scores of one configuration ({origin: scores})."""
    mase = pd.Series({origin: s['MASE'] for origin, s in per_origin.items()})
    post = mase[mase.index.isin(post_covid_origins)]
    latest = per_origin.get(LATEST_ORIGIN, {})
    return {
        'MASE_mean': float(mase.mean()),
        'MASE_worst': float(mase.max()),
        'MASE_post': float(post.mean()) if len(post) else np.nan,
        'MASE_direct_mean': float(
            np.mean([s['MASE_direct'] for s in per_origin.values()])
        ),
        'MASE_series_mean': float(
            np.mean([s['MASE_series'] for s in per_origin.values()])
        ),
        'n_origins': len(mase),
        # the latest origin forecasts the validation year (val.parquet)
        **{key: latest.get(key, np.nan) for key in METRICS},
    }

def setup_for(config, origin):
    covid_rows = effective_covid_rows(config.covid_rows, origin)
    return setups[origin, config.transform, covid_rows]

def forecast_table(config, origin, tree_params=None):
    predicted, model = FORECASTERS[config.strategy](
        config, setup_for(config, origin), val_dates_at(origin), tree_params
    )
    table = predicted.pivot_table(
        index='date', columns=TS_ID, values=TARGET, aggfunc='first'
    )[SERIES]
    if not np.all(np.isfinite(table.to_numpy())):
        raise ValueError('non-finite forecast after the inverse transform')
    return table, model

def evaluate_at(config, origin, tree_params=None):
    """Fit, forecast and score one configuration at one origin (worker process)."""
    result = {'origin': origin}
    start = time.perf_counter()

```

```

with warnings.catch_warnings():
    warnings.simplefilter('ignore')
    try:
        table, model = forecast_table(config, origin, tree_params)
        result.update(score_forecast(table, config.family, origin))
        result.update(model_info(config.family, model))
        result['error'] = None
    except Exception as exc: # noqa: BLE001 – reported in the results table
        result['error'] = f'{type(exc).__name__}: {exc}[:250]'
result['fit_seconds'] = time.perf_counter() - start
return result

def config_origins(config):
    return list(ORIGINS)

def summarize(config, origin_results, tree_params=None):
    """One results row per configuration: the origins aggregated."""
    row = {**config._asdict(), 'feature_set': set_label(config.feature_set)}
    ok = {r['origin']: r for r in origin_results if r['error'] is None}
    errors = [r['error'] for r in origin_results if r['error'] is not None]
    row['error'] = errors[0] if errors else None
    row['fit_seconds'] = sum(r['fit_seconds'] for r in origin_results)
    row['origin_mase'] = {origin: r['MASE'] for origin, r in ok.items()}
    if not errors:
        row.update(aggregate(ok))
    latest = ok.get(LATEST_ORIGIN, {})
    for key in [
        'prediction',
        'abs_errors',
        'forecast',
        'series_mase',
        'n_trees',
        'alpha',
    ]:
        if key in latest:
            row[key] = latest[key]
    edges = [r['alpha_at_edge'] for r in ok.values() if 'alpha_at_edge' in r]
    if edges:
        row['alpha_at_edge'] = f'{sum(edges)}/{len(edges)}'
    row['tuned'] = tree_params is not None
    row['tree_params'] = tree_params
    return row

example_config = Config(
    'lightgbm', 'seasonal_log_diff', 'all', 'keep', 'direct_single', FEATURE_SETS[0]
)
example = evaluate_at(example_config, LATEST_ORIGIN)
assert example['error'] is None, example['error']
print(example_config.label)
print(
    f'Validation year: bottom-up MASE {example["MASE"]:.3f} | '
    f'direct MASE {example["MASE_direct"]:.3f} | '
    f'series MASE {example["MASE_series"]:.3f}'
)

```

```
)
pd.DataFrame(
    {
        'actual total': panel_wide.loc[val_dates_at(LATEST_ORIGIN), 'total'],
        'bottom-up': example['prediction'],
        'direct': example['forecast']['total'],
    },
).round(0)
```

LightGBM | seasonal_log_diff | all | keep | direct_single | lags+rolling+domain+static+cal_fourier

Validation year: bottom-up MASE 4.165 | direct MASE 4.305 | series MASE 2.615

Out[11]:

	actual total	bottom-up	direct
--	--------------	-----------	--------

date			
2024-09-30	445389.0	368956.0	342695.0
2024-10-31	508738.0	405390.0	396175.0
2024-11-30	560732.0	500602.0	487478.0
2024-12-31	806478.0	625470.0	603237.0
2025-01-31	1483669.0	978559.0	996498.0
2025-02-28	1326884.0	837774.0	819146.0
2025-03-31	929096.0	705674.0	698434.0
2025-04-30	686239.0	391392.0	392785.0
2025-05-31	461341.0	317830.0	315741.0
2025-06-30	444882.0	311593.0	312095.0
2025-07-31	620143.0	413153.0	409147.0
2025-08-31	576049.0	395675.0	391264.0

6. Baselines

The references of notebooks 5 and 7, applied to every series at every origin:

- **naive**: every validation month repeats the last training month.
- **seasonal naive**: every validation month repeats the same month one year earlier.

The bottom-up sum of the seasonal naive forecasts is the seasonal naive forecast of the total, so its `MASE_mean` is the same as in notebooks 5 and 7.

```
In [12]: def baseline_forecast(origin, seasonal):
          history = panel_wide.loc[:origin]
          future = val_dates_at(origin)
          if seasonal:
              values = history.reindex(future - pd.offsets.MonthEnd(12)).to_numpy()
```

```

else:
    values = np.tile(history.iloc[-1].to_numpy(), (len(future), 1))
    return pd.DataFrame(values, index=future, columns=SERIES)

baseline_scores = {
    name: {o: score_forecast(baseline_forecast(o, seasonal), name, o) for o in ORIG
    for name, seasonal in [('naive', False), ('seasonal_naive', True)]
}
baselines = pd.DataFrame({
    name: aggregate(per_origin) for name, per_origin in baseline_scores.items()
}).T
BASELINE_MASE = float(baselines.loc['seasonal_naive', 'MASE_mean'])
baseline_forecasts = {
    name: scores[LATEST_ORIGIN]['prediction']
    for name, scores in baseline_scores.items()
}

display(
    pd.DataFrame({
        name: {origin_label(o): s['MASE'] for o, s in per_origin.items()}
        for name, per_origin in baseline_scores.items()
    }).rename_axis('seasonal MASE of the total by origin')
)
baselines[[*AGG_METRICS, *METRICS]].astype(float).round(3)

```

	naive	seasonal_naive
seasonal MASE of the total by origin		
2017-08	3.325103	0.680184
2018-08	2.441616	0.977196
2023-08	3.393431	1.462635
2024-08	6.148383	4.023995

Out[12]:

	MASE_mean	MASE_worst	MASE_post	MASE_direct_mean	MASE_series_me
naive	3.827	6.148	4.771	3.827	2.9
seasonal_naive	1.786	4.024	2.743	1.786	1.9

7. Grid Search

Every combination of the six choices at every origin, with the execution pattern of notebook 7:

1. **Fit in parallel.** Each (configuration, origin) task runs in a separate process with `joblib` (`N_JOBS` workers, one BLAS thread each). At the pre-COVID origins `remove` is the same model as `keep`, so that task is fitted once and shared.

2. **Aggregate** the origins of each configuration with `summarize` .

Each fit trains on the whole panel (about 20 times the rows of notebook 7) and the recursive strategy rebuilds the panel features 12 times per forecast, so the grid is smaller than notebook 7's: about 430 configurations and 1,300 fits, roughly 45–90 minutes on 8 cores.

```
In [ ]: configs = [
    Config(*choices)
    for choices in itertools.product(
        FAMILIES, TARGET_TRANSFORMS, WINDOWS, COVID_ROWS, STRATEGIES, FEATURE_SETS
    )
]

def task_key(config, origin):
    """Before the pandemic `remove` is the same model as `keep`: share the task."""
    return config._replace(
        covid_rows=effective_covid_rows(config.covid_rows, origin)
    ), origin

tasks = list(
    dict.fromkeys(
        task_key(config, origin)
        for config in configs
        for origin in config_origins(config)
    )
)
print(
    f'{len(configs)} configurations × their origins → {len(tasks)} fits '
    f'on {N_JOBS} workers...'
)

grid_start = time.perf_counter()
with parallel_config(backend='loky', inner_max_num_threads=1):
    outputs = Parallel(n_jobs=N_JOBS, verbose=5)(
        delayed(evaluate_at)(config, origin) for config, origin in tasks
    )
print(f'Fitting done in {time.perf_counter() - grid_start:,.0f}s')

by_task = dict(zip(tasks, outputs))
results = [
    summarize(config, [by_task[task_key(config, o)] for o in config_origins(config)]
    for config in configs
]
results_df = pd.DataFrame(results)
failed = results_df[results_df['error'].notna()]
print(f'Failed: {len(failed)} / {len(results_df)}')
if len(failed):
    display(failed['error'].str.split(':', 1).str[0].value_counts().rename('failures'))
```

432 configurations × their origins → 1296 fits on 8 workers...

```

[Parallel(n_jobs=8)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=8)]: Done   2 tasks      | elapsed:   2.0min
[Parallel(n_jobs=8)]: Done  56 tasks      | elapsed:   2.6min
[Parallel(n_jobs=8)]: Done 146 tasks      | elapsed:   3.4min
[Parallel(n_jobs=8)]: Done 272 tasks      | elapsed:   4.3min
[Parallel(n_jobs=8)]: Done 434 tasks      | elapsed:   5.4min
[Parallel(n_jobs=8)]: Done 632 tasks      | elapsed:   6.6min
[Parallel(n_jobs=8)]: Done 866 tasks      | elapsed:   8.6min
[Parallel(n_jobs=8)]: Done 1136 tasks     | elapsed:  12.9min
[Parallel(n_jobs=8)]: Done 1296 out of 1296 | elapsed:  15.2min finished
Fitting done in 910s
Failed: 0 / 432

```

8. Results

All configurations that produced a forecast at every origin, ranked by the mean seasonal MASE of the bottom-up total. The seasonal naive is the bar to beat.

```

In [14]: PARAM_KEYS = ['family', 'transform', 'window', 'covid_rows', 'strategy', 'feature_s

def rank(frame):
    ranked = (
        frame[frame['error'].isna()].sort_values('MASE_mean').reset_index(drop=True)
    )
    ranked['beats_seasonal_naive'] = ranked['MASE_mean'] < BASELINE_MASE
    return ranked

ranked = rank(results_df)
print(
    f'{len(ranked)} configurations ranked; '
    f'{ranked["beats_seasonal_naive"].sum()} beat the seasonal naive '
    f'(mean seasonal MASE {BASELINE_MASE:.3f})'
)
ranked[[*PARAM_KEYS, *AGG_METRICS, 'RMSE', 'MAPE', 'fit_seconds']].head(15).round(3

```

432 configurations ranked; 113 beat the seasonal naive (mean seasonal MASE 1.786)

Out[14]:

	family	transform	window	covid_rows	strategy	
0	ridge	mean_scale	all	remove	recursive	lags+rolling+domain
1	ridge	mean_scale	all	remove	recursive	lags+rolling+domain+s
2	ridge	mean_scale	all	remove	recursive	lags+growth+rolling+domain+s
3	lightgbm	seasonal_log_diff	last_10y	remove	recursive	lags+rolling+domain
4	lightgbm	seasonal_log_diff	all	remove	recursive	lags+growth+rolling+domain+s
5	xgboost	seasonal_log_diff	all	remove	recursive	lags+growth+rolling+domain+s
6	lasso	seasonal_log_diff	all	remove	recursive	lags+rolling+domain
7	lasso	seasonal_log_diff	last_10y	remove	recursive	lags+rolling+domain
8	lightgbm	seasonal_log_diff	all	remove	recursive	lags+rolling+domain
9	ridge	seasonal_log_diff	last_10y	remove	recursive	lags+rolling+domain+s
10	xgboost	seasonal_log_diff	last_10y	remove	recursive	lags+rolling+domain
11	lasso	seasonal_log_diff	all	remove	recursive	lags+growth+rolling+domain+s
12	lasso	seasonal_log_diff	all	remove	recursive	lags+rolling+domain+s
13	lightgbm	seasonal_log_diff	all	remove	recursive	lags+rolling+domain+s
14	xgboost	seasonal_log_diff	all	remove	recursive	lags+rolling+domain+s

Tuning the Tree Models (Optuna)

As in notebook 7: the **best grid configuration of every tree family × training window group** (LightGBM and XGBoost, each with `all`, `last_10y` and `last_6y`) keeps its six choices and only the tree parameters are searched (TPE sampler), with the **mean seasonal MASE of the bottom-up total over the origins** as the objective. Tuning one configuration per window shows whether a window that looked worse with the default parameters catches up once the trees are tuned for it. The first trial is the grid's default parameters. LightGBM cannot combine `linear_tree` with the L1 objective, so those trials are pruned. Ridge and Lasso are not in this search: their penalty is already chosen by temporal cross-validation.

Six studies of 25 trials (fewer than notebook 7's 40, because every trial fits the whole panel at every origin), about 15–25 minutes. The second table shows the tuned mean MASE of each family × window group.

```
In [15]: N_TRIALS = 25
optuna.logging.set_verbosity(optuna.logging.WARNING)

DEFAULT_TRIAL = {
    'lightgbm': {
        'num_leaves': 8,
        'min_child_samples': 10,
```



```

        'reg_alpha': 1e-3,
        'reg_lambda': 1e-3,
        'objective': 'regression',
        'linear_tree': False,
    },
    'xgboost': {
        'max_depth': 3,
        'min_child_weight': 3.0,
        'reg_alpha': 1e-3,
        'reg_lambda': 1.0,
        'objective': 'reg:squarederror',
    },
}

def suggest_tree_params(trial, family):
    regularization = {
        'reg_alpha': trial.suggest_float('reg_alpha', 1e-3, 10, log=True),
        'reg_lambda': trial.suggest_float('reg_lambda', 1e-3, 10, log=True),
    }
    if family == 'lightgbm':
        params = {
            'num_leaves': trial.suggest_int('num_leaves', 4, 32, log=True),
            'min_child_samples': trial.suggest_int('min_child_samples', 3, 30),
            'objective': trial.suggest_categorical(
                'objective', ['regression', 'regression_l1', 'huber']
            ),
            'linear_tree': trial.suggest_categorical('linear_tree', [False, True]),
            **regularization,
        }
        if params['linear_tree'] and params['objective'] == 'regression_l1':
            raise optuna.TrialPruned('LightGBM cannot fit linear trees with an L1 loss')
        return params
    return {
        'max_depth': trial.suggest_int('max_depth', 2, 6),
        'min_child_weight': trial.suggest_float('min_child_weight', 1, 20, log=True),
        'objective': trial.suggest_categorical(
            'objective',
            ['reg:squarederror', 'reg:absoluteerror', 'reg:pseudohubererror'],
        ),
        **regularization,
    }

def evaluate_origins(config, tree_params):
    with parallel_config(backend='loky', inner_max_num_threads=1):
        return Parallel(n_jobs=min(N_JOBS, len(ORIGINS)))(
            delayed(evaluate_at)(config, origin, tree_params)
            for origin in config_origins(config)
        )

def row_config(row):
    return Config(
        row['family'],
        row['transform'],
    )

```

```

        row['window'],
        row['covid_rows'],
        row['strategy'],
        canonical(row['feature_set'].split('+')),
    )

def tune(config):
    def objective(trial):
        tree_params = suggest_tree_params(trial, config.family)
        origin_results = evaluate_origins(config, tree_params)
        if any(r['error'] is not None for r in origin_results):
            raise optuna.TrialPruned('failed at an origin')
        return float(np.mean([r['MASE'] for r in origin_results]))

    study = optuna.create_study(
        direction='minimize', sampler=optuna.samplers.TPESampler(seed=RANDOM_STATE)
    )
    study.enqueue_trial(DEFAULT_TRIAL[config.family])
    study.optimize(objective, n_trials=N_TRIALS)
    tree_params = dict(study.best_params)
    return study, summarize(config, evaluate_origins(config, tree_params), tree_params)

tune_start = time.perf_counter()
grid_rows = ranked[~ranked['tuned']]
studies, tuned_rows = {}, []
for family in sorted(TREE_FAMILIES):
    for window in WINDOWS:
        group = grid_rows[
            (grid_rows['family'] == family) & (grid_rows['window'] == window)
        ]
        if group.empty:
            continue
        studies[family, window], tuned_row = tune(row_config(group.iloc[0]))
        tuned_rows.append(tuned_row)

print(
    f'Tuned {len(studies)} family × window groups in '
    f'{time.perf_counter() - tune_start:.0f}s'
)

results_df = pd.concat([results_df, pd.DataFrame(tuned_rows)], ignore_index=True)
ranked = rank(results_df)

tuning = pd.DataFrame([
    {
        'family': FAMILY_NAMES[family],
        'window': window,
        'configuration': row_config(tuned_row).label,
        'grid MASE_mean': study.trials[0].value,
        'tuned MASE_mean': study.best_value,
        'gain': study.trials[0].value - study.best_value,
        'trials (complete / pruned)': (
            f'{sum(t.state.name == "COMPLETE" for t in study.trials)} / '
            f'{sum(t.state.name == "PRUNED" for t in study.trials)}'
        ),
    },

```

```

        'best parameters': study.best_params,
    }
    for ((family, window), study), tuned_row in zip(studies.items(), tuned_rows)
])
display(tuning.round(3))
display(
    tuning
    .pivot_table(index='window', columns='family', values='tuned MASE_mean')
    .reindex(index=[w for w in WINDOWS if w in set(tuning['window'])])
    .rename_axis('tuned MASE_mean')
    .round(3)
)

```

Tuned 6 family × window groups in 1,056s

	family	window	configuration	grid MASE_mean	tuned MASE_mean	gain	trials (complete / pruned)	
0	LightGBM	all	LightGBM seasonal_log_diff all remove ...	1.514	1.414	0.101	25 / 0	0.01673
1	LightGBM	last_10y	LightGBM seasonal_log_diff last_10y remo...	1.499	1.478	0.021	25 / 0	0.00158
2	LightGBM	last_6y	LightGBM seasonal_log_diff last_6y keep ...	1.548	1.231	0.317	25 / 0	0.2691
3	XGBoost	all	XGBoost seasonal_log_diff all remove r...	1.493	1.415	0.078	25 / 0	0.206
4	XGBoost	last_10y	XGBoost seasonal_log_diff last_10y remov...	1.514	1.497	0.017	25 / 0	0.07
5	XGBoost	last_6y	XGBoost seasonal_log_diff last_6y keep ...	1.563	1.101	0.462	25 / 0	0.001358

	family	LightGBM	XGBoost
tuned MASE_mean			
	all	1.414	1.415
	last_10y	1.478	1.497
	last_6y	1.231	1.101

Log to MLflow

Only the configurations whose mean seasonal MASE is **below** `MLFLOW_MAX_MASE = 1.5` are logged (grid and tuned runs), one run each in the experiment **Brazil Tourism Forecasting ML Global**: the six choices and the panel as parameters; the bottom-up, direct and series MASE, the validation-year metrics, the seasonal MASE of each origin (`MASE_origin_YYYY_MM`) and the tree count or penalty as metrics (`eval_split = validation`). The rest stay in `results_df` only.

```
In [16]: MLFLOW_MAX_MASE = 1.5
N_LOG_THREADS = 8
ORIGIN_LABELS = ', '.join(origin_label(o) for o in ORIGINS)
PANEL_LABEL = (
    f'{len(COMPONENTS)} components (routes {"/".join(ROUTES.values())}, '
    f'top {N_SUBDIVISIONS} subdivisions, ≥ {MIN_MONTHLY_ARRIVALS}/month) '
    '+ remainder + total'
)

def finite(value):
    return isinstance(value, (int, float, np.number)) and np.isfinite(value)

def log_result(client, result):
    params = {
        **{k: result[k] for k in PARAM_KEYS},
        'panel': PANEL_LABEL,
        'n_series': len(SERIES),
        'horizon': HORIZON,
        'train_end': f'{TRAIN_END:%Y-%m}',
        'eval_split': 'validation',
        'origins': ORIGIN_LABELS,
        'selection_metric': 'mean_seasonal_mase_bottom_up',
        'tuned': result.get('tuned', False),
    }
    if isinstance(result.get('tree_params'), dict):
        params['tree_params'] = json.dumps(result['tree_params'])
    if isinstance(result.get('alpha_at_edge'), str):
        params['alpha_at_edge'] = result['alpha_at_edge']
    values = {
        **{key.replace(' ', '_'): result.get(key) for key in [*METRICS, *AGG_METRICS]},
        **{
            f'MASE_origin_{origin:%Y_%m}': value
            for origin, value in result['origin_mase'].items()
        },
        'n_trees': result.get('n_trees'),
        'alpha': result.get('alpha'),
        'fit_seconds': result['fit_seconds'],
    }
    timestamp = int(time.time() * 1000)
    metrics = [
        mlflow.entities.Metric(key, float(value), timestamp, 0)
        for key, value in values.items()
        if finite(value)
    ]
```

```

run_name = ' | '.join(str(result[k]) for k in PARAM_KEYS)
run = client.create_run(
    experiment.experiment_id,
    run_name=f'{run_name} | tuned' if result.get('tuned') else run_name,
)
client.log_batch(
    run.info.run_id,
    metrics=metrics,
    params=[mlflow.entities.Param(k, str(v)) for k, v in params.items()],
)
client.set_terminated(run.info.run_id, status='FINISHED')

to_log = ranked[ranked['MASE_mean'] < MLFLOW_MAX_MASE].to_dict('records')
client = mlflow.MlflowClient()
log_start = time.perf_counter()
with ThreadPoolExecutor(max_workers=N_LOG_THREADS) as pool:
    list(
        tqdm(
            pool.map(lambda r: log_result(client, r), to_log),
            total=len(to_log),
            desc='Logging to MLflow',
        )
    )
print(
    f'Logged {len(to_log)} of {len(ranked)} ranked configurations '
    f'(MASE_mean < {MLFLOW_MAX_MASE}) in {time.perf_counter() - log_start:,.0f}s'
)

```

```

Logging to MLflow: 100%|██████████| 12/12 [00:02<00:00, 5.44it/s]
Logged 12 of 438 ranked configurations (MASE_mean < 1.5) in 2s

```

Best Configuration per Family

The best configuration of each family (tuned or not) by mean seasonal MASE of the bottom-up total. The dashed line is the seasonal naive.

```

In [17]: best_per_family = ranked.groupby('family', sort=False).first()
best_per_family = best_per_family.reindex([
    f for f in FAMILIES if f in best_per_family.index
]).reset_index()
best_per_family['label'] = best_per_family['family'].map(FAMILY_NAMES) + np.where(
    best_per_family['tuned'], ' (tuned)', ''
)

BAR_METRICS = [
    'MASE_mean',
    'MASE_worst',
    'MASE_direct_mean',
    'MASE_series_mean',
    'RMSE',
    'MAPE',
]
METRIC_LABELS = {

```

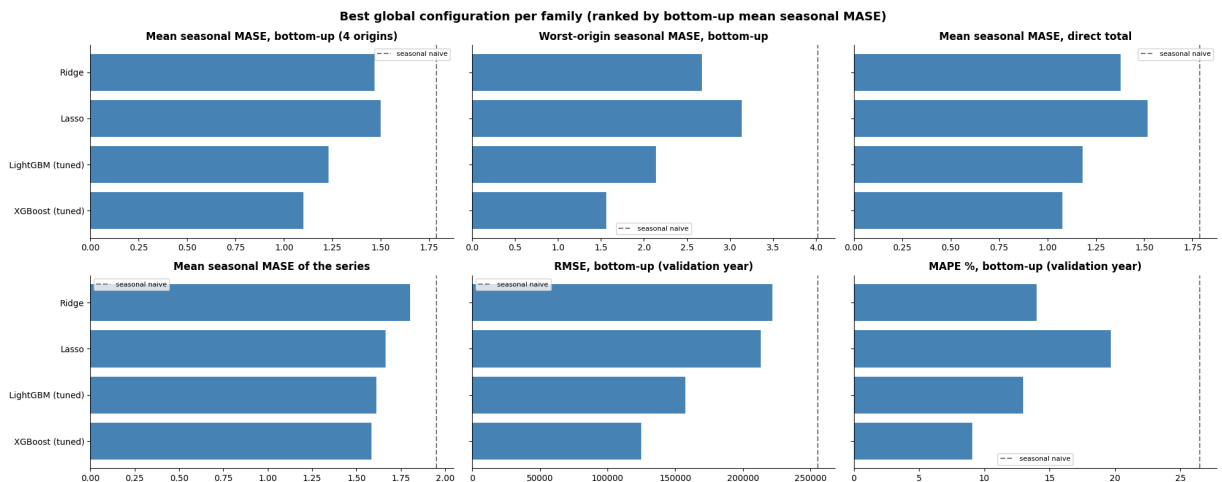
```

'MASE_mean': f'Mean seasonal MASE, bottom-up ({len(ORIGINS)} origins)',
'MASE_worst': 'Worst-origin seasonal MASE, bottom-up',
'MASE_direct_mean': 'Mean seasonal MASE, direct total',
'MASE_series_mean': 'Mean seasonal MASE of the series',
'RMSE': 'RMSE, bottom-up (validation year)',
'MAPE': 'MAPE %, bottom-up (validation year)',
}

fig, axes = plt.subplots(2, 3, figsize=(20, 8), sharey=True)
fig.suptitle(
    'Best global configuration per family (ranked by bottom-up mean seasonal MASE)'
    fontsize=14,
    fontweight='bold',
)
for ax, col in zip(axes.flat, BAR_METRICS):
    ax.barh(
        best_per_family['label'][:, :-1], best_per_family[col][:, :-1], color='steelblue'
    )
    ax.axvline(
        float(baselines.loc['seasonal_naive', col]),
        color='grey',
        linestyle='--',
        label='seasonal naive',
    )
    ax.set_title(METRIC_LABELS[col], fontweight='bold')
    ax.legend(fontsize=8)
    sns.despine(ax=ax)
plt.tight_layout()
plt.show()

best_per_family[
    [*PARAM_KEYS, 'tuned', *AGG_METRICS, *METRICS, 'n_trees', 'alpha', 'alpha_at_ed
].round(3)

```



Out[17]:

	family	transform	window	covid_rows	strategy	
0	ridge	mean_scale	all	remove	recursive	lags+rolling+domain
1	lasso	seasonal_log_diff	all	remove	recursive	lags+rolling+domain
2	lightgbm	seasonal_log_diff	last_6y	keep	recursive	lags+rolling+domain
3	xgboost	seasonal_log_diff	last_6y	keep	recursive	lags+growth+rolling+domain+st

Top 10 Configurations — Validation Forecasts of the Total

The same view as notebooks 5 and 7: the ten best configurations (by mean seasonal MASE over the origins), each with the last 24 training months of the total, the actual validation year and the **bottom-up** forecast of the latest origin. The training line is coloured by model family.

```
In [18]: FAMILY_COLORS = {
    'ridge': 'steelblue',
    'lasso': 'mediumpurple',
    'lightgbm': 'seagreen',
    'xgboost': 'darkorange',
}
TOP_N_FORECASTS = 10

top10 = ranked.head(TOP_N_FORECASTS)
val_total = panel_wide.loc[val_dates_at(LATEST_ORIGIN), 'total']
train_plot = panel_wide.loc[:TRAIN_END, 'total'].iloc[-24:]

fig, axes = plt.subplots(5, 2, figsize=(14, 26))
fig.suptitle(
    f'Top 10 Global Configurations – Validation Set Forecasts of the Total '
    f'({val_total.index[0]:%b %Y}–{val_total.index[-1]:%b %Y})',
    fontsize=15,
    fontweight='bold',
)

for i, (ax, (_, row)) in enumerate(zip(axes.flat, top10.iterrows())):
    ax.plot(
        train_plot.index,
        train_plot.values,
        label='Train (last 24 m)',
        linewidth=1.5,
        color=FAMILY_COLORS[row['family']],
        alpha=0.7,
    )
    ax.plot(
        val_total.index, val_total.values, label='Actual', linewidth=2, color='black'
    )
    ax.plot(
        val_total.index,
        row['prediction'],
```

```

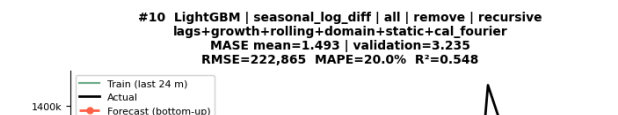
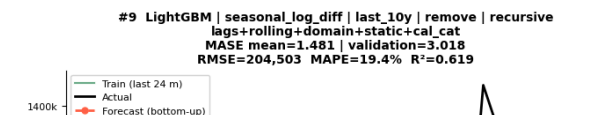
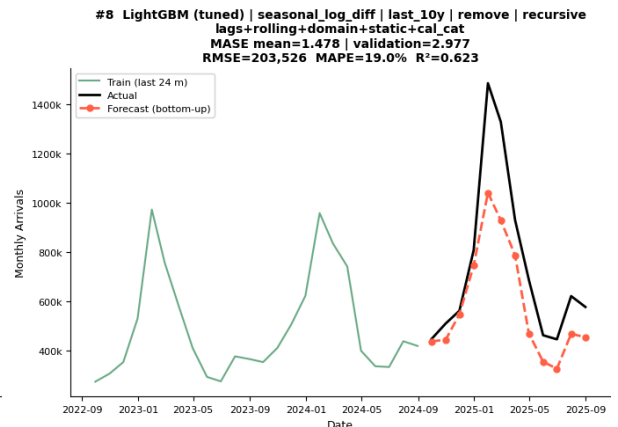
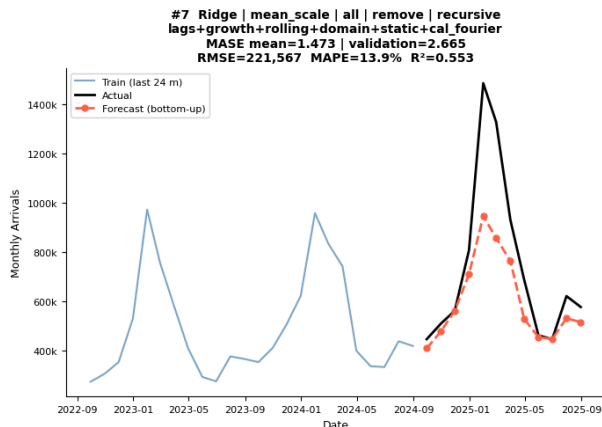
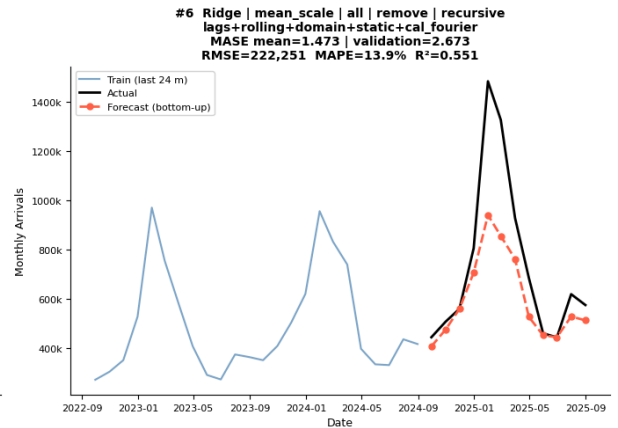
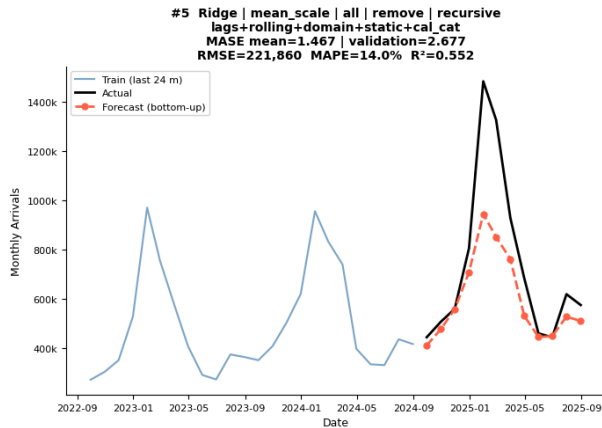
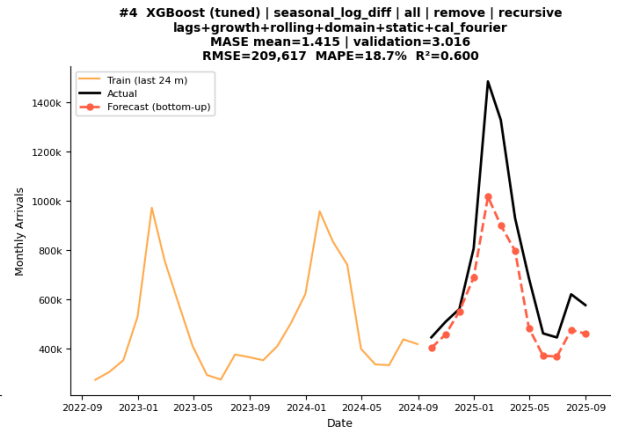
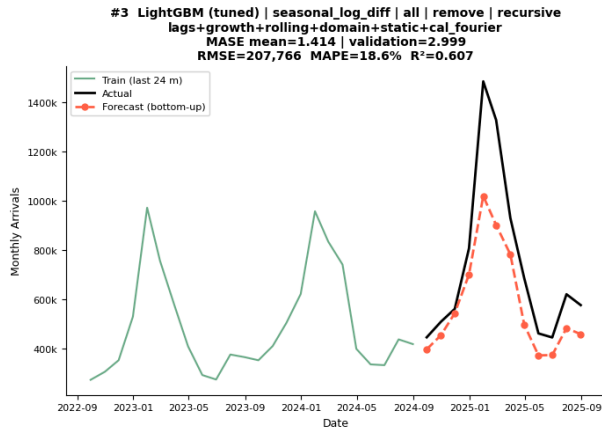
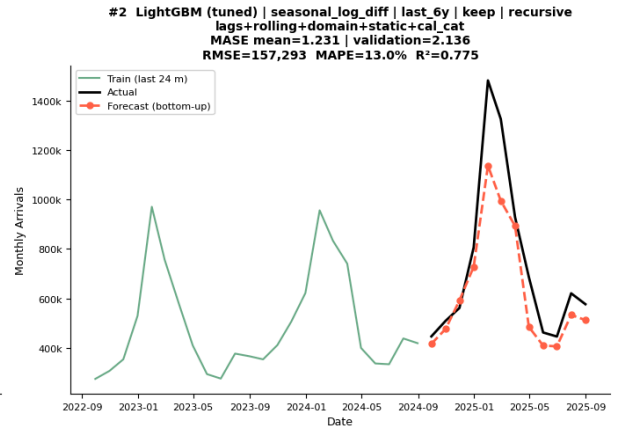
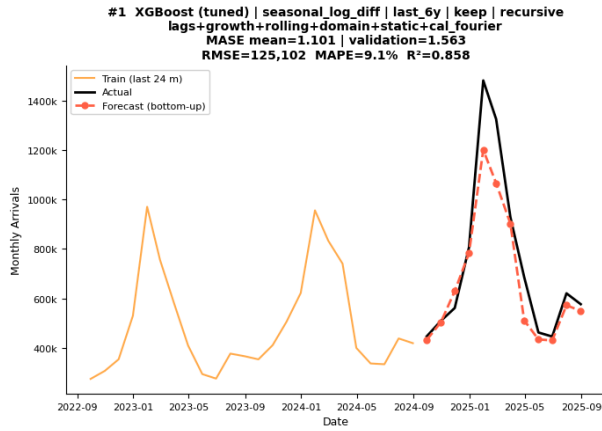
        label='Forecast (bottom-up)',
        linewidth=2,
        linestyle='--',
        color='tomato',
        marker='o',
        markersize=5,
    )
    ax.set_title(
        f'#{i + 1} {FAMILY_NAMES[row["family"]]} '
        f'" (tuned)" if row["tuned"] else "" | '
        f'{row["transform"]} | '
        f'{row["window"]} | {row["covid_rows"]} | {row["strategy"]}\n'
        f'{row["feature_set"]}\n'
        f'MASE mean={row["MASE_mean"]:.3f} | validation={row["MASE"]:.3f}\n'
        f'RMSE={row["RMSE"]:.0f} MAPE={row["MAPE"]:.1f}% R2= {row["R2"]:.3f}',
        fontsize=10,
        fontweight='bold',
    )
    ax.set_xlabel('Date', fontsize=9)
    ax.set_ylabel('Monthly Arrivals', fontsize=9)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _ : f'{int(x / 1000)}k')
    ax.tick_params(axis='both', labelsize=8)
    ax.legend(fontsize=8)
    sns.despine(ax=ax)

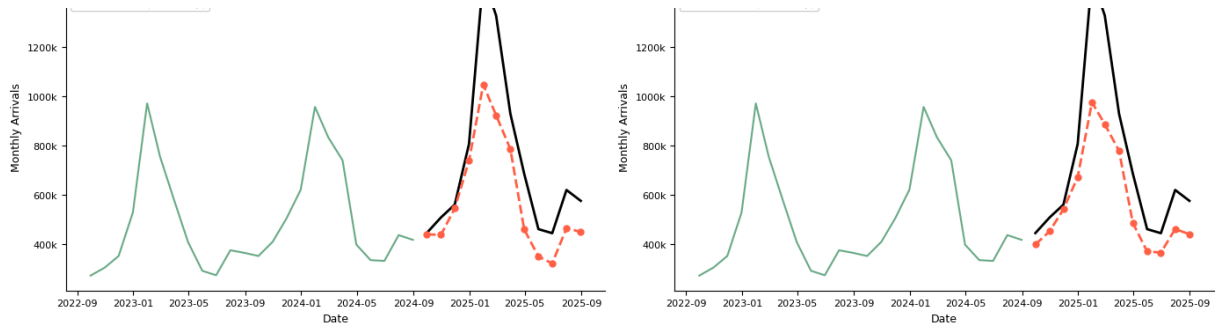
for ax in axes.flat[len(top10) :]:
    ax.set_visible(False)

plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()

```


Top 10 Global Configurations — Validation Set Forecasts of the Total (Sep 2024-Aug 2025)



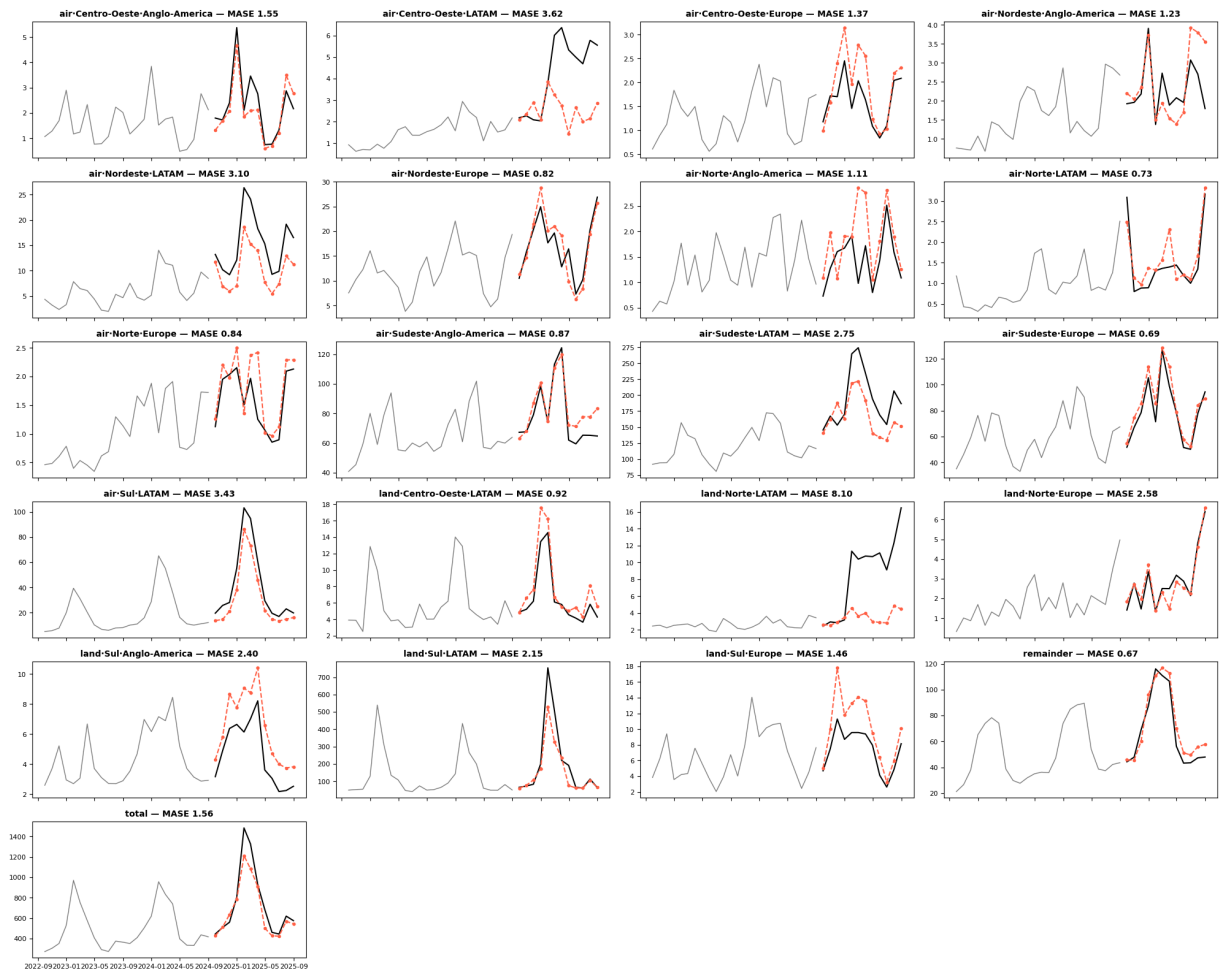


The Best Configuration, Series by Series

The validation-year forecast of every series by the best configuration (latest origin), with the seasonal MASE of each series. It shows where the bottom-up error comes from.

```
In [19]: best = ranked.iloc[0]
fig, axes = plt.subplots(n_rows, N_COLS, figsize=(20, 2.8 * n_rows), sharex=True)
for ax, series_id in zip(axes.flat, SERIES):
    history = panel_wide.loc[:TRAIN_END, series_id].iloc[-24:]
    actual = panel_wide.loc[val_total.index, series_id]
    ax.plot(history.index, history / 1e3, color='grey', linewidth=1)
    ax.plot(actual.index, actual / 1e3, color='black', linewidth=1.5)
    ax.plot(
        actual.index,
        best['forecast'][series_id] / 1e3,
        color='tomato',
        linestyle='--',
        marker='o',
        markersize=3,
    )
    ax.set_title(
        f'{series_id} - MASE {best["series_mase"][series_id]:.2f}',
        fontsize=10,
        fontweight='bold',
    )
    ax.tick_params(labelsize=8)
for ax in axes.flat[len(SERIES) :]:
    ax.set_visible(False)
fig.suptitle(
    f'Best configuration: {row_config(best).label} - arrivals (thousands); '
    'black = actual, red = forecast',
    fontsize=13,
    fontweight='bold',
)
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()
```

Best configuration: XGBoost | seasonal_log_diff | last_6y | keep | recursive | lags+growth+rolling+domain+static+cal_fourier — arrivals (thousands); black = actual, red = forecast



9. Heatmaps of the Combinations

As in notebooks 5 and 7, each cell summarizes all grid configurations that share the two choices on its axes with the **p95 of the mean seasonal MASE** (the 5% quantile of the cell, the value that only the best 5% of its configurations beat), and the number under it is how many configurations the cell summarizes. The tuned runs are left out.

- Green is better. The colour scale stops at $1.5 \times$ the seasonal naive.
- With about 30–70 configurations per cell, the p95 is close to the best values of the cell.

```
In [20]: ORDERS = {
    'family': FAMILIES,
    'transform': TARGET_TRANSFORMS,
    'window': list(WINDOWS),
    'covid_rows': COVID_ROWS,
    'strategy': STRATEGIES,
    'feature_set': [set_label(s) for s in FEATURE_SETS],
}
HEATMAP_LAYOUT = [
    # (rows, columns, grid position)
    ('family', 'transform', (0, slice(0, 2))),
    ('family', 'strategy', (0, slice(2, 4))),
```

```

    ('family', 'covid_rows', (0, slice(4, 6))),
    ('window', 'family', (1, slice(0, 3))),
    ('feature_set', 'family', (1, slice(3, 6))),
]
MASE_LABEL_CAP = 10
P95_QUANTILE = 0.05 # p95: the value only the best 5% of a cell's configurations b
grid_ranked = ranked[~ranked['tuned']]

def p95(values):
    return values.quantile(P95_QUANTILE)

def mase_label(value, count):
    if pd.isna(value):
        return ''
    shown = f'>{MASE_LABEL_CAP}' if value > MASE_LABEL_CAP else f'{value:.2f}'
    return f'{shown}\n(n={int(count)})'

def p95_mase_pivot(rows, columns):
    """p95 of the mean seasonal MASE per cell, and the configurations per cell."""
    frames = [
        grid_ranked.pivot_table(
            index=rows, columns=columns, values='MASE_mean', aggfunc=aggfunc
        ).reindex(
            index=[v for v in ORDERS[rows] if v in grid_ranked[rows].unique()],
            columns=[v for v in ORDERS[columns] if v in grid_ranked[columns].unique()]
        )
        for aggfunc in (p95, 'count')
    ]
    if rows == 'family':
        frames = [frame.rename(index=FAMILY_NAMES) for frame in frames]
    if columns == 'family':
        frames = [frame.rename(columns=FAMILY_NAMES) for frame in frames]
    pivot, counts = frames
    labels = pd.DataFrame(
        [
            [mase_label(v, n) for v, n in zip(values, ns)]
            for values, ns in zip(pivot.to_numpy(), counts.to_numpy())
        ],
        index=pivot.index,
        columns=pivot.columns,
    )
    return pivot, labels

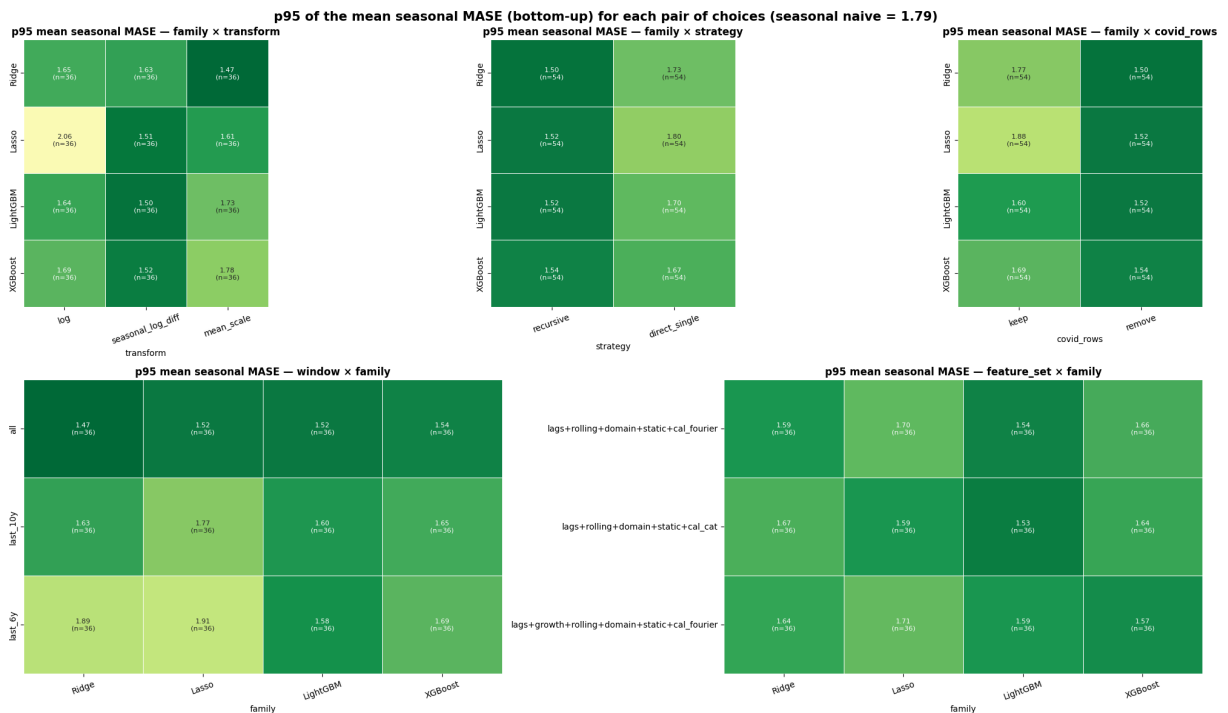
fig = plt.figure(figsize=(20, 12))
grid = fig.add_gridspec(2, 6, height_ratios=[1, 1.1])
for rows, columns, (r, c) in HEATMAP_LAYOUT:
    ax = fig.add_subplot(grid[r, c])
    pivot, labels = p95_mase_pivot(rows, columns)
    sns.heatmap(
        pivot,
        annot=labels,
        fmt='',

```

```

annot_kws={'fontsize': 8},
cmap='RdYlGn_r',
vmin=grid_ranked['MASE_mean'].min(),
vmax=BASELINE_MASE * 1.5,
linewidths=0.4,
cbar=False,
ax=ax,
)
ax.set_title(f'p95 mean seasonal MASE - {rows} x {columns}', fontweight='bold')
ax.set_xlabel(columns)
ax.set_ylabel('')
ax.tick_params(axis='x', labelrotation=20)
fig.suptitle(
    'p95 of the mean seasonal MASE (bottom-up) for each pair of choices '
    f'(seasonal naive = {BASELINE_MASE:.2f})',
    fontsize=15,
    fontweight='bold',
)
plt.tight_layout()
plt.show()

```



11. Feature Importance of the Best Configuration per Family

The best configuration of each family is refitted at the latest origin (with its tuned parameters if it was tuned):

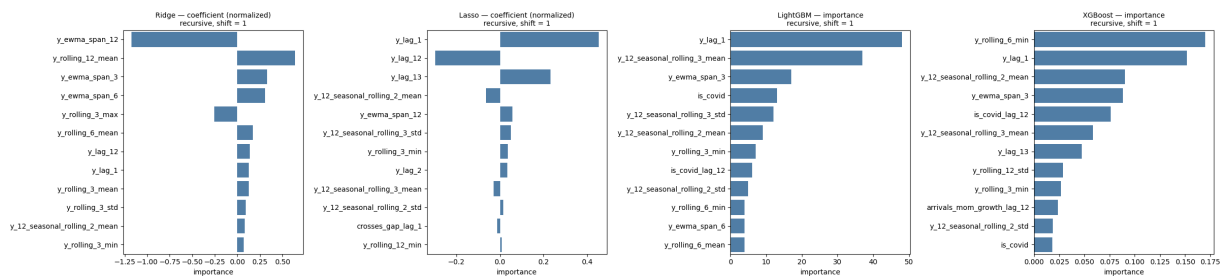
- **Native importance** (`MLForecast.feature_importance()`): normalized coefficients for Ridge and Lasso, built-in importance for the trees.
- **Permutation importance by group** on the validation year: all the columns of a group are shuffled together across the panel rows, the model predicts again, and the increase

of the MAE of the **bottom-up total** is measured. The features are built from the actual validation history (teacher forcing). The `static` group shows how much the model relies on knowing which series it is forecasting.

```
In [21]: def refit(row):
    config = row_config(row)
    tree_params = row['tree_params'] if isinstance(row['tree_params'], dict) else None
    setup = setup_for(config, LATEST_ORIGIN)
    shift = 1 if config.strategy == 'recursive' else HORIZON
    frame = with_future_rows(setup['history'], val_dates_at(LATEST_ORIGIN))
    with warnings.catch_warnings():
        warnings.simplefilter('ignore')
        model, feature_config, _ = fit_for_shift(
            config, setup, frame, shift, tree_params
        )
    return config, shift, model, feature_config

best_models = {row['family']: refit(row) for _, row in best_per_family.iterrows()}

TOP_FEATURES = 12
fig, axes = plt.subplots(1, len(best_models), figsize=(24, 5.5))
for ax, family in zip(np.atleast_1d(axes), best_models):
    config, shift, model, _ = best_models[family]
    importance = model.feature_importance().head(TOP_FEATURES)
    sns.barplot(data=importance, x='importance', y='feature', ax=ax, color='steelblue')
    kind = 'importance' if family in TREE_FAMILIES else 'coefficient (normalized)'
    ax.set_title(
        f'{FAMILY_NAMES[family]} - {kind}\n{config.strategy}, shift = {shift}',
        fontsize=10,
    )
    ax.set_ylabel('')
plt.tight_layout()
plt.show()
```



```
In [22]: N_REPEATS = 20

def group_permutation_importance( # noqa: PLR0913, PLR0914, PLR0917
    config, shift, model, feature_config, n_repeats=N_REPEATS
):
    """Bottom-up validation MAE increase when the columns of a group are shuffled."""
    setup = setup_for(config, LATEST_ORIGIN)
    val_months = val_dates_at(LATEST_ORIGIN)
    observed = pd.concat(
        [setup['history'], panel[panel['date'].isin(val_months)]], ignore_index=True
```

```

)
for series_id, transformer in setup['transformers'].items():
    is_val = (observed[TS_ID] == series_id) & observed['date'].isin(val_months)
    values = observed.loc[observed[TS_ID] == series_id].set_index('date')[TARGET]
    if isinstance(transformer, SeasonalLog1pDiffTarget):
        z = np.log1p(values) - np.log1p(values).shift(SEASONAL_PERIOD)
    else:
        z = transformer.transform(values)
    observed.loc[is_val, Y] = z.loc[val_months].to_numpy()
features, groups = build_features(observed, shift)
rows = features[features['date'].isin(val_months)].reset_index(drop=True)
actual = panel_wide.loc[val_months, 'total'].to_numpy()

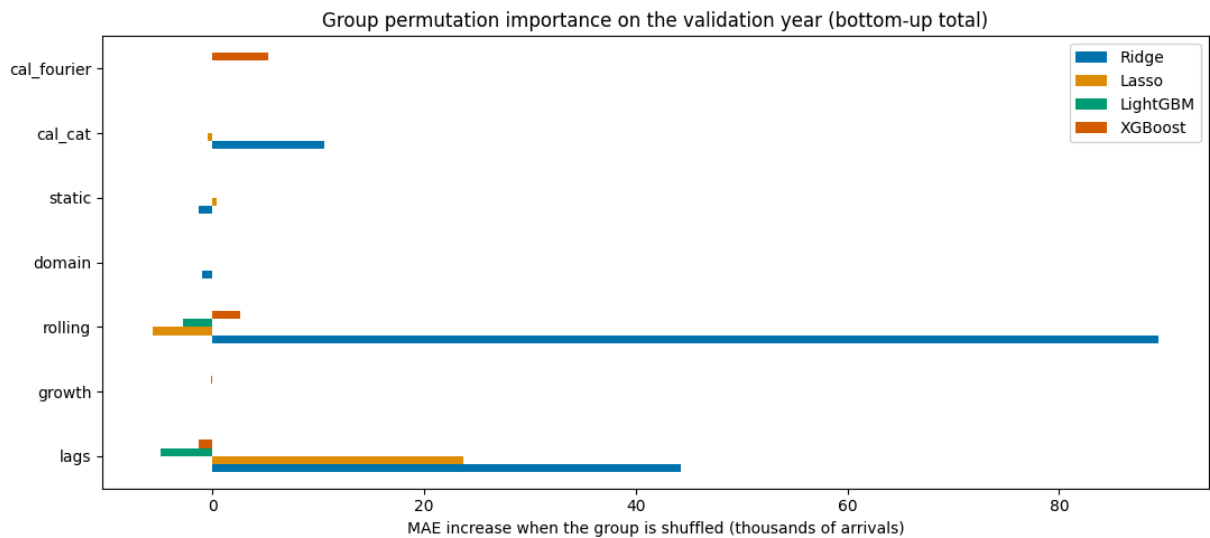
def mae_of(frame):
    predicted = to_arrivals(setup, predicted_frame(model, feature_config, frame))
    table = predicted.pivot_table(
        index='date', columns=TS_ID, values=TARGET, aggfunc='first'
    )
    return float(np.mean(np.abs(actual - table[BOTTOM_UP].sum(axis=1)).to_numpy()))

base = mae_of(rows)
rng = np.random.default_rng(RANDOM_STATE)
increases = {}
for group in config.feature_set:
    scores = []
    for _ in range(n_repeats):
        shuffled = rows.copy()
        order = rng.permutation(len(rows))
        shuffled[groups[group]] = (
            rows[groups[group]].iloc[order].set_axis(rows.index)
        )
        scores.append(mae_of(shuffled) - base)
    increases[group] = np.mean(scores)
return pd.Series(increases)

with warnings.catch_warnings():
    warnings.simplefilter('ignore')
    permutation = (
        pd
        .DataFrame({
            FAMILY_NAMES[family]: group_permutation_importance(*best_models[family])
            for family in best_models
        })
        .reindex(GROUP_ORDER)
        .dropna(how='all')
    )

ax = (permutation / 1e3).plot.barh(figsize=(11, 5))
ax.set_title('Group permutation importance on the validation year (bottom-up total)')
ax.set_xlabel('MAE increase when the group is shuffled (thousands of arrivals)')
plt.tight_layout()
plt.show()
permutation.round(0)

```



Out[22]:

	Ridge	Lasso	LightGBM	XGBoost
lags	44261.0	23740.0	-4941.0	-1295.0
growth	NaN	NaN	NaN	-180.0
rolling	89398.0	-5662.0	-2825.0	2569.0
domain	-934.0	-16.0	0.0	0.0
static	-1274.0	440.0	0.0	-32.0
cal_cat	10584.0	-405.0	0.0	NaN
cal_fourier	NaN	NaN	NaN	5311.0

12. Save the Best Configuration's Validation Forecasts

The ensembling notebook (10) combines the best configuration of every approach, so it needs their **forecasts**, not only their metrics. The best configuration of this notebook (rank 1 by mean seasonal MASE, tuned or not) is re-run at every validation origin with its tuned tree parameters if it was tuned (the fits are deterministic), and its forecasts are logged as the artifact `validation_forecasts.csv` of a run tagged `role = best_validation_forecasts` in this notebook's experiment:

Column	Content
<code>origin</code>	the validation origin (YYYY-MM)
<code>date</code>	the forecast month
<code>actual</code>	the actual total arrivals
<code>forecast</code>	the forecast of the total: the bottom-up sum of the components and the remainder, the view the ranking uses

The mean seasonal MASE of these forecasts must match the ranking, which is checked before logging. The run has no `eval_split` parameter, so the comparison queries of the other notebooks do not pick it up as a grid run.

```
In [23]: best = ranked.iloc[0]
best_config = row_config(best)
best_tree_params = (
    best['tree_params'] if isinstance(best['tree_params'], dict) else None
)
origin_results = evaluate_origins(best_config, best_tree_params)
assert all(r['error'] is None for r in origin_results)

best_forecasts = pd.concat(
    [
        pd.DataFrame({
            'origin': origin_label(r['origin']),
            'date': val_dates_at(r['origin']),
            'actual': panel_wide.loc[val_dates_at(r['origin']), 'total'].to_numpy(),
            'forecast': r['prediction'], # bottom-up total
        })
        for r in origin_results
    ],
    ignore_index=True,
)

mase_mean = float(np.mean([r['MASE'] for r in origin_results]))
assert np.isclose(mase_mean, best['MASE_mean']), (mase_mean, best['MASE_mean'])

def log_validation_forecasts(approach, configuration, forecasts, mase_mean):
    """Log the best configuration's forecasts at every origin as an MLflow artifact
    with mlflow.start_run(
        run_name=f'BEST validation forecasts | {configuration}':250
    ):
        mlflow.set_tags({'role': 'best_validation_forecasts', 'approach': approach})
        mlflow.log_params({
            'configuration': configuration[:500],
            'origins': ', '.join(sorted(forecasts['origin'].unique())),
            'horizon': HORIZON,
        })
        mlflow.log_metric('MASE_mean', mase_mean)
        mlflow.log_text(forecasts.to_csv(index=False), 'validation_forecasts.csv')
        print(f'{approach}: logged {len(forecasts)} forecasts of {configuration}')

label = best_config.label + (' | tuned' if best['tuned'] else '')
log_validation_forecasts('Global ML', label, best_forecasts, mase_mean)
```

Global ML: logged 48 forecasts of XGBoost | seasonal_log_diff | last_6y | keep | recursive | lags+growth+rolling+domain+static+cal_fourier | tuned

Reading the Results

Question	Where
Does a global model beat the seasonal naive on the total?	<code>beats_seasonal_naive</code> in section 8, the dashed lines of the bar charts
Is the bottom-up total better than the global model's own forecast of the total?	<code>MASE_mean</code> vs <code>MASE_direct_mean</code> (best per family)
Does the global approach beat the local models of notebooks 5 and 7?	Section 10
Which target processing suits a global model?	Heatmap <i>family</i> × <i>transform</i>
Does removing the pandemic years help across the panel?	Heatmap <i>family</i> × <i>covid_rows</i>
Where does the bottom-up error come from?	The series-by-series chart of the best configuration
Does the model need to know which series it forecasts?	The <code>static</code> bar of the permutation importance

Things to keep in mind:

- **Selection on validation data.** As in notebooks 5 and 7, the best of hundreds of configurations is optimistic; compare through the heatmaps and `MASE_worst`, and keep the test year for the final notebook.
- **The panel is fixed from the training data.** The series list uses the last training year (volume threshold) for every origin; it is a design choice of the panel, not a fitted parameter.
- **Remainder.** The remainder series mixes everything that is not a kept component. If it dominates the bottom-up error, a finer panel (more components) is the next step.